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**32384 CHARLES COUNTY POMONKEY PUMP STATION
IMPROVEMENTS**

SECONDARY CONTROL PANEL RESUBMITTAL

SECTION 17441 - CONTROLS SYSTEM HARDWARE

March 05, 2026

ENGINEER: WRA

CONTRACTOR: JOHNSTON CONSTRUCTION

***MANUFACTURERS' REPRESENTATIVE FOR ENGINEERED INSTRUMENTATION,
PROCESS EQUIPMENT AND PUMPING SYSTEMS***

Branch Offices: RICHMOND, VA and LEOLA, PA

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

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32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

General Comments:

Resubmittal Comment Confirmations:

1. Panel Size: Per Contract Drawing I-8, backplate width shall be 11ft. Submitted panel widths are as follows:
 - SCP: 20.00 in.
 - SPMP-01: 31.38 in.
 - SPMP-02: 31.38 in.
 - SFAN-01: 16.00 in.

Based on Contract Drawings, UPS and PLC panels are estimated to be 20 and 30 inches wide, respectively. Total width of all panels currently comes to 149in. approximately, which will not suit the 11ft backplate. Contractor shall minimize panel widths (to match Contract Drawings) as necessary to meet backplate width requirements.

Sherwood Logan Response:

The width of the RVSS panel enclosures have been modified so that all of the submitted panel dimensions should fit on the 11ft backplate.

2. Charles County Comment - High Level Float: It appears the high-level float will activate the secondary control mode. We want to make sure the high-level alarm does not generate repeatedly when the station is in secondary mode.

Sherwood Logan Response:

The high-level float does activate the secondary control mode. According to the PLC I/O List in Spec Section 17600 3.07 the Secondary Control Panel High Level Alarm displays, but is not a PCS alarm. Thus, the repetitive alarm

concern should not present an issue.

Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION
1	SCE	SCE-24H2010SS6LP	Wall Mount Enclosure - NEMA 4X
1	SCE	SCE-24P20	Enclosure Backpanel
2	Schneider Electric	9001SKT38LWW31	PTT Light, 120VAC LED, White
2	Schneider Electric	9001SKT38LYA31	PTT Light, 120VAC LED, Amber
3	Schneider Electric	9001SKR1B	Pushbutton, Black
3	Schneider Electric	9001KA3	1 NC Contact
4	Weidmuller	7940102575	Circuit Breaker, 120VAC, 5A
4	Weidmuller	7940125379	Circuit Breaker, 120VAC, 1A
5	Schneider Electric	RXM2AB2F7	Miniature Relay, 120VAC, 8 Blade
5	Schneider Electric	RXZE2M114	Miniature Relay Socket, 8 Blade, DIN
6	Schneider Electric	RXM3AB2F7	Miniature Relay, 120VAC, 11 Blade
6	Schneider Electric	RXZE2S111M	Miniature Relay Socket, 11 Blade, DIN
7	Square D	9050JCK16V20	Timing Relay, 0-10 min, 120VAC, 8 Pin Octal, DPDT
7	Square D	8501NR51	Timing Relay Socket, 8 Pin, DIN
8	Diversified	ISO-120-AAE	Isolated Switch, 2 Channel
9	Eaton	D851LA	Alternating Relay
9	Eaton	D3PA2	Alternating Relay Socket
10	SCE	SCE-TSH25	Enclosure Heater
10	Finder	7T.81.0.000.2403	Enclosure Heater Thermostat, NC
11	Phoenix Contact	3044076	Terminal Block
11	Phoenix Contact	3044092	Grounding Terminal Block
11	Phoenix Contact	3047028	Terminal End Barrier
11	Phoenix Contact	0800886	Terminal Anchor

**32384 CHARLES COUNTY POMONKEY PUMP STATION
IMPROVEMENTS**

Data Sheet 1

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: SCE: Wall Mount Enclosure - NEMA 4X Model Number: SCE-24H2010SS6LP Tags/Service: ENC / Secondary Control Panel Specifications: • 0.075" stainless stell type 316/316L • Flange trough collar around all sides of door opening • Pour in place oil & water resistant gasket • Stainless stell continuous hinges • Hasp and staple for padlocking
1	Mfg: SCE: Enclosure Backpanel Model Number: SCE-24P20 Tags/Service: ENC / Secondary Control Panel Specifications: • 21"H x 17"W x .88"D • Powder coated white finish



Saginaw Control and Engineering
95 Midland Road Saginaw, MI 48638-5770
(800) 234-6871 - Fax: (989) 799-4524
SCE@SaginawControl.com

WALL MOUNT ENCLOSURE - NEMA 4X

SCE-24H2010SS6LP

TAGS: ENC

SCE-24EL2010SS6LP - Cost Effective Alternative.



Product Specifications:

Part Number: SCE-24H2010SS6LP
Description: Nema 4X LP Enclosure
Height: 24.00"
Width: 20.00"
Depth: 10.00"
Price Code: S3
List Price: \$1,652.52
Catalog Page: 270
Est. Ship Weight: 49.00 lbs

Construction

- * 0.075 In. stainless steel Type 316/316L.
- * Continuously welded seams ground smooth.
- * Flange trough collar around all sides of door opening.
- * Pour in place oil & water resistant gasket
- * Collar studs 3/8-16 provided for mounting optional subpanels.
- * Hasp and staple for padlocking.
- * Stainless steel continuous hinges.
- * Removable print pocket furnished if height and width of enclosure is greater than 12 Inches.
- * Ground stud on door.
- * Stainless steel clamps on three sides door when greater than 30 inch wide.
- * Welded mounting provision top and bottom of back.

Application

Designed to house electrical and electronic controls, instruments and components in areas which may be regularly hosed down or are very wet conditions. Provides protection from dust, dirt, oil and water. For outdoor application a drip shield and drain vent is recommended.

For details about the design, performance expectations, applications and design suggestions - See Design Considerations
www.saginawcontrol.com/instman/considerations.pdf

Finish

#4 brushed finish on all exterior surfaces. Optional sub-panels are powder coated white.

Industry Standards - (IS6)

- * NEMA Type 3R, 4, 4X, 12 and Type 13
- * UL Listed Type 3R, 4, 4X and 12
- * CSA Type 4, 4X and 12
- * IEC 60529
- * IP 66

Notes

Provision for Lifting Lugs on enclosures when Height >48" and Depth >16". The Lifting Lug assembly will be included with the enclosure bolt pack.

Special Instructions apply for IS3, IS4 and IS6 to maintain the environmental rating of Type 3R for these parts. Instructions are located on the enclosure door. Drip shield is required on IS3, drip shield is recommended on IS4 and IS6. Drain holes are required on all.

Optional Accessories

SCE-24P20 Subpanel, Bent
SCE-24P20GALV Subpanel, Bent Galvanized
SCE-BVK Breather Vent
SCE-DS20SS Shield, S.S. Drip
SCE-LPCLAMPSS Assembly, S.S. LP Clamp (2pcs)
SCE-LPQR LP Style Quick Release Clamp Assy (Stainless)
SCE-RD24H20SS6 Door, Replacement

Similar Part Numbers

SCE-16H1206SS6LP Nema 4X LP Enclosure
SCE-16H1208SS6LP Nema 4X LP Enclosure
SCE-16H1606SS6LP Nema 4X LP Enclosure
SCE-16H2006SS6LP Nema 4X LP Enclosure
SCE-20H1606SS6LP Nema 4X LP Enclosure
SCE-20H1608SS6LP Nema 4X LP Enclosure
SCE-20H1610SS6LP Nema 4X LP Enclosure
SCE-20H2006SS6LP Nema 4X LP Enclosure
SCE-20H2408SS6LP Nema 4X LP Enclosure
SCE-24H1608SS6LP Nema 4X LP Enclosure

Installation Information

- * Drip Shield Kit Assembly
- * Swing Panel Assembly for LP Enclosures
- * Clamp Fastened Single Door Enclosures (Stainless Steel)
- * NEMA 12 and 4 Clamp Fastened Single Door Enclosure



Saginaw Control and Engineering
 95 Midland Road Saginaw, MI 48638-5770
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 SCE@SaginawControl.com

ENCLOSURE BACKPANEL

SCE-24P20

TAGS: ENC

Product Specifications:



Part Number: SCE-24P20
Description: Subpanel, Bent
Height: 21.00"
Width: 17.00"
Depth: 0.88"
Price Code: P3
List Price: \$56.29
Catalog Page: 444
Est. Ship Weight: 14.00 lbs

Finish

Powder Coated White.

Industry Standards - (IS17)

- ⚙ NEMA Not Applicable
- ⚙ UL Not Applicable
- ⚙ CSA N/A

Similar Part Numbers

~~SCE-10P10Subpanel, Flat~~
~~SCE-10P6Subpanel, Flat~~
~~SCE-10P8Subpanel, Flat~~
~~SCE-12DLP12Subpanel, Flat~~
~~SCE-12P10Subpanel, Flat~~
~~SCE-12P12Subpanel, Flat~~
~~SCE-12P12CSubpanel, Flat~~
~~SCE-12P16CSubpanel, Flat~~
~~SCE-12P20CSubpanel, Flat~~
~~SCE-12P24Subpanel, Bent~~

Installation Information

- ⚙ Sub-Plate Layout & Grounding for 10-32

Data Sheet 2

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Schneider Electric: PTT Light, 120VAC LED, White Model Number: 9001SKT38LWW31 Tags/Service: LT4203 / Secondary Control Panel Specifications: • Fresnel plastic lens type • Full voltage 120V light block supply • Plastic bezel material • White lens color • White LED • Round signaling unit head • Polycarbonate lens material
2	Mfg: Schneider Electric: PTT Light, 120VAC LED, Amber Model Number: 9001SKT38LYA31 Tags/Service: LT4307, LT4311 / Secondary Control Panel Specifications: • Fresnel plastic lens type • Full voltage 120V light block supply • Plastic bezel material • Amber lens color • Yellow LED • Round signaling unit head

	• Polycarbonate lens material
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Product data sheet

Specifications

PTT LIGHT, 120VAC LED, WHITE



Push-to-test pilot light, Harmony 9001SK, plastic, polycarbonate, fresnel lens, white, 30mm, LED white, 120V

TAGS: LT4203

9001SKT38LWW31

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony 9001SK
Product or Component Type	Push-to-test pilot light
Lens type	Fresnel plastic
Light block supply	Full voltage 120 V

Complementary

Bezel material	Plastic
Mounting diameter	1.2 in (30 mm)
Shape of signaling unit head	Round
Cap/operator or lens colour	White
Light source	White LED
Bulb base	BA 9s
Operating Mode	Push-to-test
Lens type	Fresnel
[Us] rated supply voltage	120 V
Light source colour	White
Contacts type and composition	1 C/O
Lens material	Polycarbonate
Size	30 mm
Light source	LED

Environment

Protective treatment	TC
IP Degree of Protection	IP65 IP66

Ordering and shipping details

Category	US10CS121429
Discount Schedule	0CS1
GTIN	785901546443
Returnability	Yes
Country of origin	MX

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	6.350 in (16.129 cm)
Package 1 Width	2.50 in (6.35 cm)
Package 1 Length	16.129 in (40.968 cm)
Package weight(Lbs)	4.160 oz (117.934 g)

Contractual warranty

Warranty (in months)	18
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Product data sheet

Specifications

PTT LIGHT, 120VAC LED, AMBER



Push-to-test pilot light, Harmony 9001SK, plastic, polycarbonate, fresnel lens, amber, 30mm, LED yellow, 120V

TAGS: LT4307, LT4311

9001SKT38LYA31

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony 9001SK
Product or Component Type	Push-to-test pilot light
Device short name	9001SK
Type of operator	spring return
Operator profile	Amber projecting, unmarked

Complementary

Bezel material	Plastic
Mounting diameter	1.2 in (30 mm)
Shape of signaling unit head	Octagonal
Light source	LED
Light source	LED
Bulb base	BA 9s
Light block supply	Direct
[Us] rated supply voltage	120 V AC/DC
[Us] rated supply voltage	120 V AC/DC
Contacts type and composition	1 C/O
Connections - terminals	Screw clamp terminals 1 x 0.22...2 x 1.5 mm ² IEC 60947-1
Tightening torque	7.08 lbf.in (0.8 N.m) IEC 60947-1
shape of screw head	Cross slotted
Mechanical durability	5000000 cycles
Operating position	Any position
[Ie] rated operational current	0.55 A 125 V, DC-13, A600-Q600 NEMA
[Ui] rated insulation voltage	250 V 3)IEC 60947-1
[Uimp] rated impulse withstand voltage	2.5 kV IEC 60947-1
Contacts material	Silver alloy contacts
Positive opening	Without
Short-circuit protection	10 A cartridge fuse IEC 60947-5-1
[Ith] conventional free air thermal current	10 A

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[Icm] rated short-circuit making capacity	12 kA 600 V AC-15, 7200 VA 30 kA 480 V AC-15, 7200 VA 60 kA 120 V AC-15, 7200 VA
Rated breaking capacity	1.2 kA 600 V AC-15, 720 VA 3 kA 480 V AC-15, 720 VA 6 kA 120 V AC-15, 720 VA 0.27 kA 250 V DC-13 0.55 kA 125 V DC-13
Net Weight	0.430 lb(US) (0.195 kg)
Device presentation	Complete product
Cap/operator or lens colour	Amber
Lens type	Fresnel
[Us] rated supply voltage	120 V
Light source colour	Yellow
Lens material	Polycarbonate
Size	30 mm

Environment

Standards	JIS C8201-5-1 CSA C22.2 No 14 IEC 60947-5-1 IEC 60947-5-4 IEC 60947-1 UL 508 JIS C 852 JIS C8201-1
Product Certifications	NEMA UL 508
Protective treatment	TC
Ambient Air Temperature for Storage	-40...158 °F (-40...70 °C)
Ambient Air Temperature for Operation	-13...158 °F (-25...70 °C)
Vibration resistance	7 gn (f= 2...500 Hz) conforming to IEC 60068-2-6
Shock resistance	50 gn IEC 60068-2-27
Electrical shock protection class	Class II conforming to IEC 61140
IP Degree of Protection	IP65 IP66
NEMA degree of protection	NEMA 1 NEMA 12 NEMA 13 NEMA 2 NEMA 3 NEMA 3R NEMA 4 NEMA 4X

Ordering and shipping details

Category	US10CS121429
Discount Schedule	0CS1
GTIN	785901500582
Returnability	Yes
Country of origin	MX

Packing Units

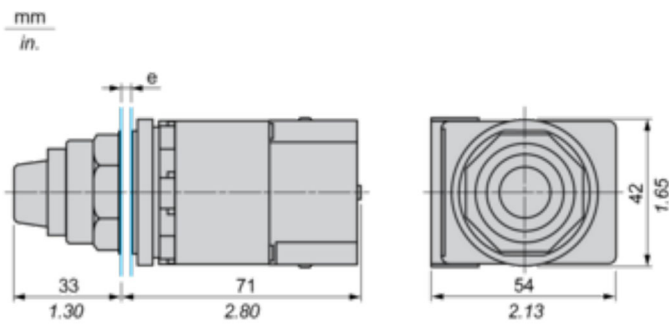
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.969 in (5.000 cm)
Package 1 Width	2.362 in (6.000 cm)
Package 1 Length	4.488 in (11.400 cm)
Package weight(Lbs)	4.233 oz (120.000 g)
Unit Type of Package 2	S01
Number of Units in Package 2	12
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	5.906 in (15.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	3.616 lb(US) (1.640 kg)
Unit Type of Package 3	P12
Number of Units in Package 3	384
Package 3 Height	17.717 in (45.000 cm)
Package 3 Width	31.496 in (80.000 cm)
Package 3 Length	47.244 in (120.000 cm)
Package 3 Weight	142.154 lb(US) (64.480 kg)

Contractual warranty

Warranty (in months)	18
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Dimensions Drawings

Dimensions

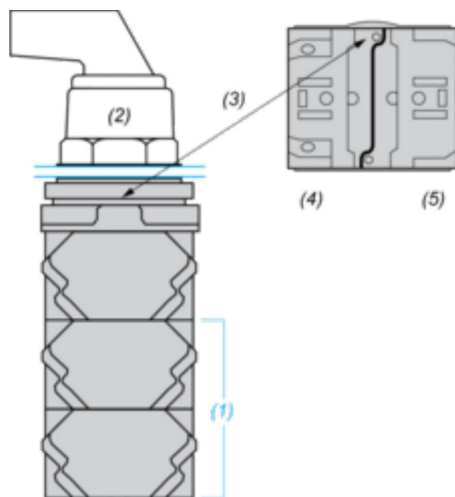


e panel thickness: 1 to 6 mm / 0.04 to 0.24 in.

Mounting and Clearance

Contact Block Mounting Position

Top and Rear views



(1) It is possible to mount up to 3 levels of contacts blocks (maximum of 6 contacts blocks)

(2) Operator

(3) Locating notch

(4) Side 1

(5) Side 2

Data Sheet 3

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Schneider Electric: Pushbutton, Black Model Number: 9001SKR1B Tags/Service: PB4306 / Secondary Control Panel Specifications: • Plastic bezel material • Octagonal signaling unit head • NEMA 4X degree of protection • Black operator color • Flush operator profile • Spring return operator • 30mm size
1	Mfg: Schneider Electric: 1 NC Contact Model Number: 9001KA3 Tags/Service: PB4306 / Secondary Control Panel Specifications: • Positived opening contact operation • Screw clamp terminals • 7.08 lbf.in tightening torque • Silver alloy contact material • 10A conventional free air thermal current • 0.55A rated operational current

	• 60kA rated short-circuit making capacity • 6kA rated breaking capacity
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Product data sheet

Specifications

PUSHBUTTON, BLACK



Push button head, Harmony
9001SK, plastic, flush, black, 30mm,
spring return

TAGS: PB4306

9001SKR1B

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony 9001SK
Product or Component Type	Head for push-button
Device short name	9001SK

Complementary

Bezel material	Plastic
Mounting diameter	1.2 in (30 mm)
Shape of signaling unit head	Octagonal
Light source	without
Light block supply	Without supply
Contacts type and composition	Without contact block
Product Certifications	CE CSA LR24590 class 3211 03 UL listed file E42259 CCN NKCR
NEMA degree of protection	NEMA 3R/4X/13
Cap/operator or lens colour	Black
Operator profile	Flush
Type of operator	Spring return
Size	30 mm

Environment

IP Degree of Protection	IP65 IP66
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Ordering and shipping details

Category	US10CS121429
Discount Schedule	0CS1
GTIN	785901793878
Returnability	Yes
Country of origin	MX

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1

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Package 1 Height	2.800 in (7.112 cm)
Package 1 Width	2.300 in (5.842 cm)
Package 1 Length	2.00 in (5.08 cm)
Package weight(Lbs)	1.600 oz (45.359 g)

Contractual warranty

Warranty (in months)	18
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30mm Push Button, Types K, SK or KX, contact block, 1 NC

9001KA3

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony 9001K Harmony 9001SK
Product or Component Type	Contact block
Device short name	9001K
Contacts type and composition	1 NC

Complementary

Terminals description ISO n°1	(3-4)NO
Net Weight	0.051 lb(US) (0.023 kg)
Contact operation	Positive opening
Positive opening	With IEC 60947-5-2
Connections - terminals	Screw clamp terminals, 1 x 0.22...2 x 2.5 mm² IEC 60947-1
Tightening torque	7.08 lbf.in (0.8 N.m) IEC 60947-1
Shape of screw head	Cross slotted
Contacts material	Silver alloy contacts
Short-circuit protection	10 A cartridge fuse IEC 60947-5-1
[Ith] conventional free air thermal current	10 A
[Ui] rated insulation voltage	250 V 3)IEC 60947-1
[Uimp] rated impulse withstand voltage	2.5 kV IEC 60947-1
[Ie] rated operational current	3 A 240 V, AC-15, A600-Q600 NEMA 0.55 A 125 V, DC-13, A600-Q600 NEMA
[Icm] rated short-circuit making capacity	12 kA 600 V AC-15, 7200 VA 45 kA 480 V AC-15, 7200 VA 30 kA 240 V AC-15, 7200 VA 60 kA 120 V AC-15, 7200 VA 0.55 kA 125 V DC-13 0.1 kA 600 V DC-13 0.27 kA 250 V DC-13
Rated breaking capacity	3 kA 240 V AC-15, 720 VA 6 kA 120 V AC-15, 720 VA 1.2 kA 600 V AC-15, 720 VA 1.5 kA 480 V AC-15, 720 VA 0.1 kA 600 V DC-13 0.27 kA 250 V DC-13 0.55 kA 125 V DC-13
[Us] rated supply voltage	0...600 V
Connections - Terminals	Screw-clamp terminals

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Environment

Protective treatment	TC
Ambient Air Temperature for Storage	-40...158 °F (-40...70 °C)
Ambient Air Temperature for Operation	-13...158 °F (-25...70 °C)
Electrical shock protection class	Class II conforming to IEC 61140
Standards	UL 508 JIS C 4520 JIS C 852 IEC 60947-5-4 CSA C22.2 No 14 IEC 60947-1 IEC 60947-5-1 UL 60947-5-1
Product Certifications	NEMA UL 508
Vibration resistance	7 gn (f= 2...500 Hz) conforming to IEC 60068-2-6
Shock resistance	50 gn IEC 60068-2-27

Ordering and shipping details

Category	US10CS121434
Discount Schedule	0CS1
GTIN	785901880028
Returnability	Yes
Country of origin	MX

Packing Units

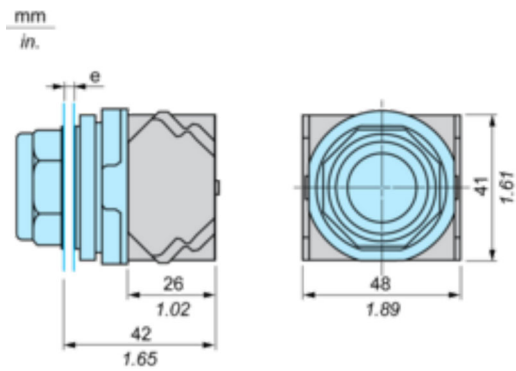
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.378 in (3.500 cm)
Package 1 Width	1.654 in (4.200 cm)
Package 1 Length	2.008 in (5.100 cm)
Package weight(Lbs)	0.811 oz (23.000 g)
Unit Type of Package 2	S01
Number of Units in Package 2	100
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	5.906 in (15.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	5.196 lb(US) (2.357 kg)

Contractual warranty

Warranty (in months)	18
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Dimensions Drawings

Dimensions

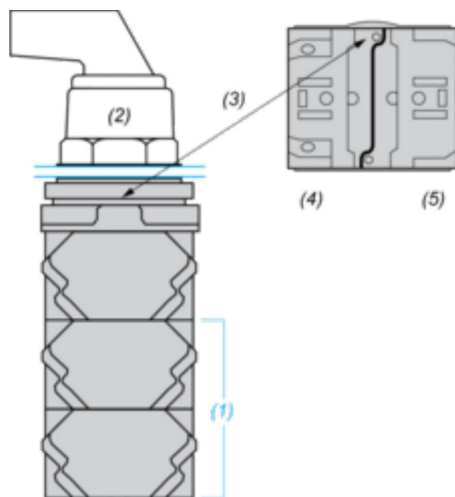


e panel thickness: 1 to 6 mm / 0.04 to 0.24 in.

Mounting and Clearance

Contact Block Mounting Position

Top and Rear views



(1) It is possible to mount up to 3 levels of contacts blocks (maximum of 6 contacts blocks)

(2) Operator

(3) Locating notch

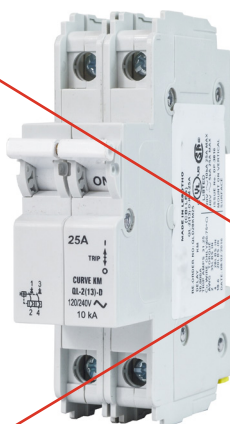
(4) Side 1

(5) Side 2

Data Sheet 4

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Weidmuller: Circuit Breaker, 120VAC, 5A Model Number: 7940102575 Tags/Service: CB4202 / Secondary Control Panel Specifications: • 120VAC operational voltage • Hydraulic magnetic type circuit breaker • 5A current rating • 10kA interrupting capacity • 100MΩ insulation resistance • 10000 mechanical operations operating life • -40...65° C operating temperature • 20in.-lb tightening torque
1	Mfg: Weidmuller: Circuit Breaker, 120VAC, 1A Model Number: 7940125379 Tags/Service: CB4202A / Secondary Control Panel Specifications: • 120VAC operational voltage • Hydraulic magnetic type circuit breaker • 1A current rating • 10kA interrupting capacity • 100MΩ insulation resistance

	• 10000 mechanical operations operating life • −40...65° C operating temperature • 20in.-lb tightening torque
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Datasheet**QL Series Hydraulic-Magnetic Miniature Circuit Breakers****Dual mount - 1 pole****DIN mount - 2 pole****Dual mount + auxiliary****Features**

- AC Circuit Breaker
- Hydraulic-Magnetic technology
- 100% rating capability, independent of ambient temperature
- One, two and three pole (UL). One, two, three pole and 1+N, 3+N (VDE).
- VDE and CCC approved, CE certified
- UL 489 Listed, CSA 22.2 certified
- Ratings 0.5A to 25A
- Optional Auxiliary Switch / Trip Alarm - factory fitted (6.5 mm wide)
- Wide range of time delays and operating currents
- Precision tripping characteristics
- Ultra compact – 13 mm wide module
- Trip indication with mid-trip handle
- Reset immediately after overload
- DIN mount product in grey shells
- Dual mount product in black shells
- Ring terminal ready (terminal screw is removable)

Applications:

- AC branch circuit protection (UL 489, IEC / EN 60847-2)
- ~~Telecom / datacom equipment~~
- ~~Lighting control~~
- ~~UPS equipment~~
- ~~Alternative energy equipment~~
- ~~Mobile power generation equipment~~
- ~~Power conditioning equipment~~

Auxiliary Switch, Trip Alarm & Combo Features

- Factory fitted
- Attached to right hand side of Circuit Breaker
- Compact 6.5 mm width
- AC or DC voltages
- UL 489 listed (Auxiliary Switch: 6 A 240VAC, 0.5 A 110VDC)
- IEC 60947-5-1 approved (Auxiliary Switch: 6 A 240 V AC, 0.5 A 110 V DC; Trip Alarm: 6 A 240 V AC, 0.5 A 110 V DC)

Optional Accessories

- Auxiliary Switch (DIN and Dual Mount)
- Auxiliary Switch + Trip Alarm (Dual Mount only)
- Trip Alarm (Dual Mount only)
- Handle lock
- Surface mounting clips
- Busbar
- 57 mm Escutcheon blank (Dual Mount only)
- 57 mm Safety blank (Dual Mount only)

Approvals

Technical data

Approvals		IEC / EN 60947-2, VDE			
Number of poles	1	2 (1+N)	2	3	4(3+N)
Operating voltages	240 V AC	240 V AC	415 V AC	415 V AC	415 V AC
Minimum current rating	0.5 A				
Maximum current rating	25 A				
Interrupting capacity	6 kA				
Operating temperature range	-40°C to +85°C				
Mounting options	DIN Rail, Dual mounting (DIN & Mini), surface mounting clip				
Time delay curves	1, 3, 9, KM, OP				

Approvals		UL 489			
Number of poles	1	1	2	3	
Operating voltages	120 V AC	240 V AC	120 / 240 V AC	240 V AC	
Minimum current rating	0.5 A				
Maximum current rating	25 A	25 A	25 A	25 A	
Ampere interrupting capacity (AIC)	10 kA	5 kA	10 kA	5 kA	
Operating temperature range	-40°C to +85°C				
Mounting options	DIN Rail, Dual mounting (DIN & Mini), surface mounting clip				
Time delay curves	1, 3, 9, KM, OP				

Insulation coordination and general data	
Endurance	10000 operations; 1500 electrical at rated current, 8500 without current (IEC 60947-2 Clause 7.2.4.2)* 6000 electrical operations (UL 489 Clause 7.1.9)*
Dielectric strength	1000-2000 Vac for one minute (IEC 60947-2 Clause 8.3.3.3)* 1000 Vac plus twice the rated voltage for one minute (UL 489 Clause 7.1.9)*
Rated impulsive withstand voltage	4 kV (IEC 60947-2)
Weight	102 g per pole, 160 g with auxiliary (unpacked)
Altitude	Certification tests done at altitude = 2000 metres. Will operate at higher altitudes.
Shock	8 G (IEC 60068-2-6) (sinusoidal wave)
Vibration	0.5 G (IEC 60068-2-6) (sinusoidal wave)
Flammability	I3 - Ignition does not persist at 850°C after glow wire is withdrawn with an oxygen index of ≥ 28
Toxicity	F1 - Smoke index of ≤ 20 which determines the fumes class
Pollution degree	PD2 - Normally only non-conductive pollution occurs. Temporary conductivity caused by condensation is to be expected.

Breaker: QL	Wire Size mm² (IEC)	Wire Gauge (UL)	Torque (IEC)	Torque (UL)	Comments
Main terminals	0.75 - 35 mm²	18 - 3 - AWG	2.5 Nm	20 in-lb	Pozidriv #2 Combi head

Long code

Example code: QL-AT-3-13-D-KM-15-R

Group	1	2	3	4	5	6	7	8	9	10
Requirement	QL Frame	Switch / Neutral	Auxiliary + Trip Alarm	Triple pole	13 mm module width	DIN Mounting	Medium delay curve KM	Current Rating 15 A	No Shunt Trip	Ring Terminal
Long Code	QL	-	AT	3	13	DM	KM	15.0	-	R

Group 1: Frame Type	Code	Description	Comments	
	QL	13 mm wide Miniature Circuit Breaker	UL 489, IEC / EN 60947-2, VDE, CE, CCC	
Group 2: Switch/Neutral	Code	Description	Comments	
	-	Not applicable	Overload poles do not have any further coding	
	N	Neutral	Green Handle	
Group 3: Auxiliary	Code	Description	Comments	
	-	Not applicable	Use this code if no Auxiliary used	
	A	Auxiliary Switch (1 x Aux in 1 module)	6.5 mm module fitted on right hand side (DIN & Dual Mount)	
	T	Trip Alarm (1 x Trip Alarm in 1 module)	6.5 mm module fitted on right hand side (Dual Mount only)	
	AT	Auxiliary Switch + Trip Alarm combo (combined in 1 module)	6.5 mm module fitted on right hand side (Dual Mount only)	
Group 4: No of Poles	Code	Description	Comments	
	1	Single pole		
	2	Double pole	2 pole or 1+N	
	3	Triple pole		
	4	Four pole	3+N only	
Group 5: Module Width	Code	Description	Comments	
	(13)	13 mm module width	13 mm per pole	
Group 6: Mounting	Code	Description	Comments	
	D	DIN rail mount	DIN mount supplied in grey only - 45 mm Escutcheon, grey body	
	DM	Dual mount	Dual mount supplied in black only - 57 mm Escutcheon, black body	
Group 7: Time Delays	Code	Description	Instantaneous Trip Point (x In)	Comments
	1	Long time delay, high instantaneous trip	10 - 20	Orange handle
	3	Short time delay	3 - 5	White handle
	9	Long time delay	7 - 12	White handle
	KM	Medium time delay	6 - 12	White handle
	OP	Instantaneous	None	White handle
Group 8: Current Ratings	Code	Description	Comments	
		0.5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 16, 20, 25 A	Ratings available vary depending on certification * Other ratings are available as special orders. Check availability.	
Group 9: Shunt Trip (not certified, only offered as special order)	Code	Description	Comments	
	-	Not applicable	If no shunt trip is required	
	U0	12 Vdc Shunt trip (Flying Leads)	Flying leads (+/- 60 mm long)	
	U1	24 Vdc Shunt trip (Flying Leads)	Recommended pulse duration 100 ms to 60 s	
	U2	48 Vdc Shunt trip (Flying Leads)	Note: Shunt trip carries no approvals - available as a special offer only	
	U3	60 Vdc Shunt trip (Flying Leads)		
	U4	110 Vdc & 110 - 230 Vac Shunt Trip (Fly Leads)		
Group 10: Special Termination	Code	Description	Comments	
	R	Ring Terminal ready	Terminal Screw is removable. Ring Lug (5 mm)	

QL Series Single Pole



Type: Single Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
0.5	QL-1-13-DM-KM-0.5	7940102571
1	QL-1-13-DM-KM-1.0	7940125379
2	QL-1-13-DM-KM-2.0	7940102572
3	QL-1-13-DM-KM-3.0	7940102573
4	QL-1-13-DM-KM-4.0	7940102574
5	QL-1-13-DM-KM-5.0	7940102575
6	QL-1-13-DM-KM-6.0	7940102576
7	QL-1-13-DM-KM-7.0	7940102577
8	QL-1-13-DM-KM-8.0	7940102578
9	QL-1-13-DM-KM-9.0	7940102579
10	QL-1-13-DM-KM-10.0	7940102580
13	QL-1-13-DM-KM-13.0	7940102581
15	QL-1-13-DM-KM-15.0	7940102582
16	QL-1-13-DM-KM-16.0	7940102583
20	QL-1-13-DM-KM-20.0	7940102584
25	QL-1-13-DM-KM-25.0	7940102585

QL Series Single Pole
(long time delay/high instant trip)

Type: Single Pole (up to 240 VAC) (long time delay/high instant trip)		
Current Ratings (amps)	Description	Order Number
1	QL-1-13-DM-1-1.0	7940115210
2	QL-1-13-DM-1-2.0	7940112653
3	QL-1-13-DM-1-3.0	7940112583
4	QL-1-13-DM-1-4.0	7940115284
5	QL-1-13-DM-1-5.0	7940113986
6	QL-1-13-DM-1-6.0	7940115211
7	QL-1-13-DM-1-7.0	7940114950
8	QL-1-13-DM-1-8.0	7940115282
10	QL-1-13-DM-1-10.0	7940102587
15	QL-1-13-DM-1-15.0	7940113987
16	QL-1-13-DM-1-16.0	7940114409
20	QL-1-13-DM-1-20.0	7940102588
25	QL-1-13-DM-1-25.0	7940112654

QL Series Double Pole



Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
0.1	QL-2-13-DM-KM-0.1	7940115220
0.5	QL-2-13-DM-KM-0.5	9926252000
1	QL-2-13-DM-KM-1.0	7940102643
2	QL-2-13-DM-KM-2.0	7940102644
3	QL-2-13-DM-KM-3.0	7940102645
4	QL-2-13-DM-KM-4.0	7940102646
5	QL-2-13-DM-KM-5.0	7940102647
6	QL-2-13-DM-KM-6.0	9926252006
7	QL-2-13-DM-KM-7.0	7940102649
8	QL-2-13-DM-KM-8.0	7940102650
10	QL-2-13-DM-KM-10.0	9926252010
12	QL-2-13-DM-KM-12.0	7940115938
13	QL-2-13-DM-KM-13.0	7940102652
15	QL-2-13-DM-KM-15.0	9926252015
16	QL-2-13-DM-KM-16.0	7940102654
20	QL-2-13-DM-KM-20.0	7940126749
25	QL-2-13-DM-KM-25.0	7940102656

QL Series Double Pole
(long time delay/high instant trip)

Type: Double Pole (up to 240 VAC) (long time delay/high instant trip)		
Current Ratings (amps)	Description	Order Number
2	QL-2-13-DM-1-2.0	7940102657
3	QL-2-13-DM-1-3.0	7940114949
4	QL-2-13-DM-1-4.0	7940113820
5	QL-2-13-DM-1-5.0	7940102658
6	QL-2-13-DM-1-6.0	7940102659
7	QL-2-13-DM-1-7.0	7940114948
10	QL-2-13-DM-1-10.0	7940102660
13	QL-2-13-DM-1-13.0	7940113821
15	QL-2-13-DM-1-15.0	7940102661
20	QL-2-13-DM-1-20.0	7940113822
25	QL-2-13-DM-1-25.0	7940113823

QL Series Triple Pole



Type: Triple Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-3(13)-DM-KM-1.0	7940102691
2	QL-3(13)-DM-KM-2.0	7940102692
5	QL-3-13-DM-KM-5.0	7940102693
10	QL-3-13-DM-KM-10.0	7940104412
15	QL-3-13-DM-KM-15.0	7940102694
20	QL-3-13-DM-KM-20.0	7940102695
25	QL-3-13-DM-KM-25.0	7940102696

QL Series Single Pole w/ Auxiliary Contact



Type: Single Pole (up to 120 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-A-1-13-DM-KM-1.0	7940102708
2	QL-A-1-13-DM-KM-2.0	7940102709
3	QL-A-1-13-DM-KM-3.0	7940102710
5	QL-A-1-13-DM-KM-5.0	7940102711
10	QL-A-1-13-DM-KM-10.0	7940102712
15	QL-A-1-13-DM-KM-15.0	7940102713
20	QL-A-1-13-DM-KM-20.0	7940102714
25	QL-A-1-13-DM-KM-25.0	7940102715

QL Series Double Pole w/ Auxiliary Contact



Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-A-2-13-DM-KM-1.0	7940114041
2	QL-A-2-13-DM-KM-2.0	7940119528
5	QL-A-2-13-DM-KM-5.0	7940119829
15	QL-A-2-13-DM-KM-15.0	7940119830
20	QL-A-2-13-DM-KM-20.0	7940119831

QL Series Single Pole (Alarm Contact)



Type: Single Pole (up to 120 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-T-1-13-DM-KM-1.0	7940114042
2	QL-T-1-13-DM-KM-2.0	7940114043
5	QL-T-1-13-DM-KM-5.0	7940114044
10	QL-T-1-13-DM-KM-10.0	7940102724
15	QL-T-1-13-DM-KM-15.0	7940114045
20	QL-T-1-13-DM-KM-20.0	7940114046
25	QL-T-1-13-DM-KM-25.0	7940114047

QL Series Double Pole (Alarm Contact)



Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-T-2-13-DM-KM-1.0	7940114048
2	QL-T-2-13-DM-KM-2.0	7940114049
5	QL-T-2-13-DM-KM-5.0	7940114050
10	QL-T-2-13-DM-KM-10.0	7940114051
15	QL-T-2-13-DM-KM-15.0	7940114052
20	QL-T-2-13-DM-KM-20.0	7940114053
25	QL-T-2-13-DM-KM-25.0	7940114054

QL Series Triple Pole (Alarm Contact)



Type: Triple Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-T-3(13)-DM-KM-1.0	7940115448
2	QL-T-3(13)-DM-KM-2.0	7940115449
5	QL-T-3(13)-DM-KM-5.0	7940115450
10	QL-T-3(13)-DM-KM-10.0	7940115451
15	QL-T-3(13)-DM-KM-15.0	7940115452
20	QL-T-3(13)-DM-KM-20.0	7940115453
25	QL-T-3(13)-DM-KM-25.0	7940115454

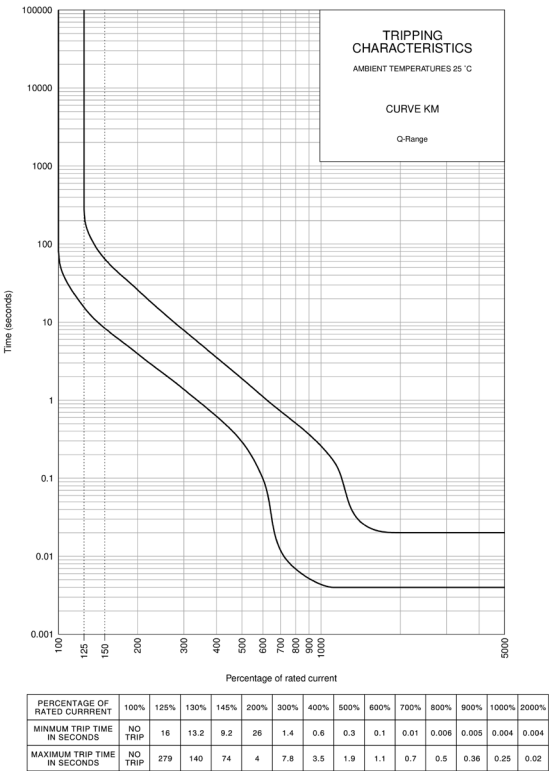
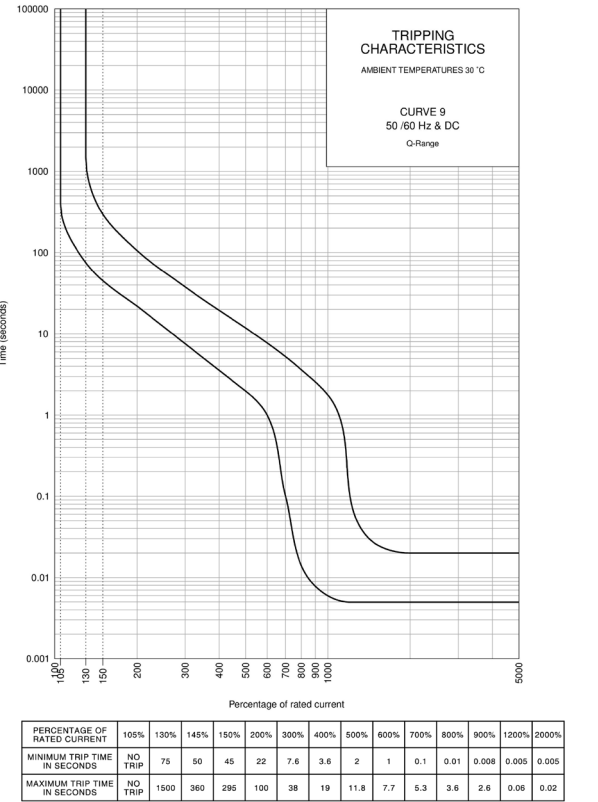
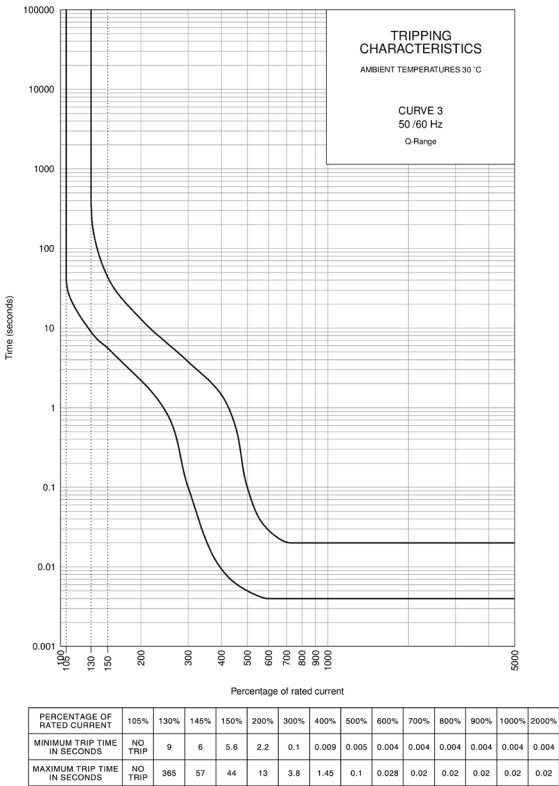
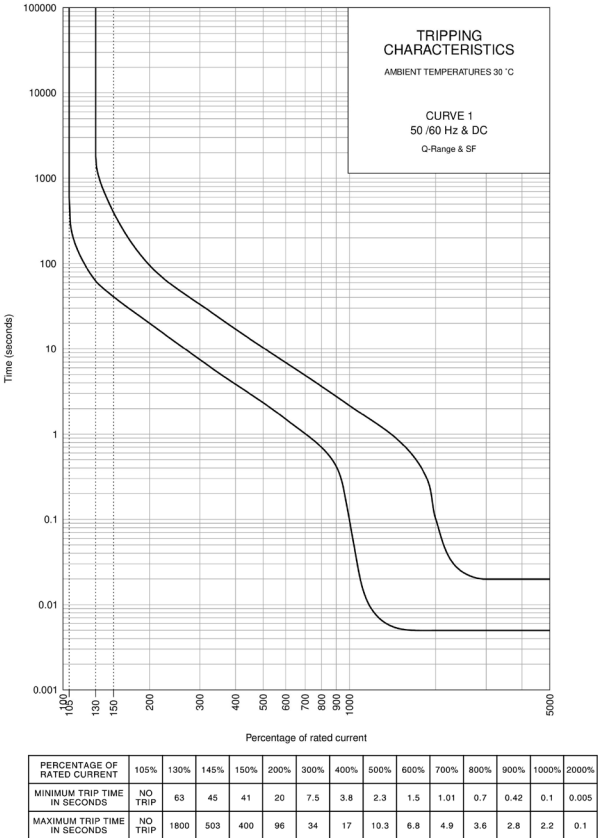
QL Series Single Pole Combination
(Auxiliary & Alarm Contact)

Type: Single Pole (up to 120 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-AT-1-13-DM-KM-1.0	7940114055
5	QL-AT-1-13-DM-KM-5.0	7940102725
10	QL-AT-1-13-DM-KM-10.0	7940102726
15	QL-AT-1-13-DM-KM-15.0	2853970000
20	QL-AT-1-13-DM-KM-20.0	7940114056
25	QL-AT-1-13-DM-KM-25.0	7940114057

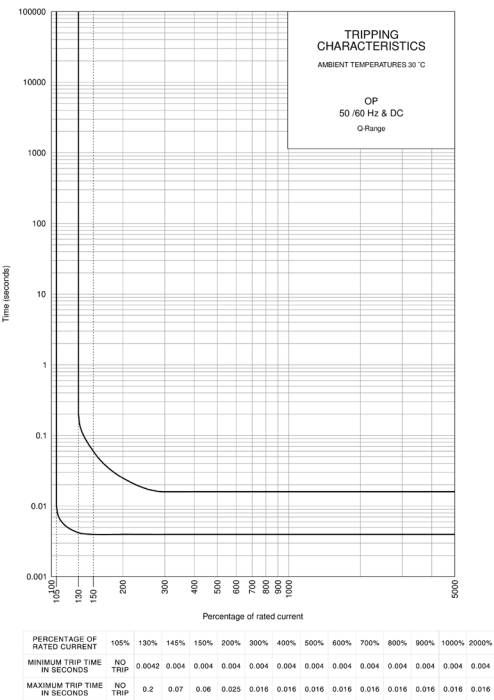
QL Series Double Pole Combination
(Auxiliary & Alarm Contact)

Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-AT-2-13-DM-KM-1.0	7940114058
2	QL-AT-2-13-DM-KM-2.0	7940114059
10	QL-AT-2-13-DM-KM-10.0	7940126754
15	QL-AT-2-13-DM-KM-15.0	7940102727
20	QL-AT-2-13-DM-KM-20.0	7940114060
25	QL-AT-2-13-DM-KM-25.0	7940114061

Time Delay Curves

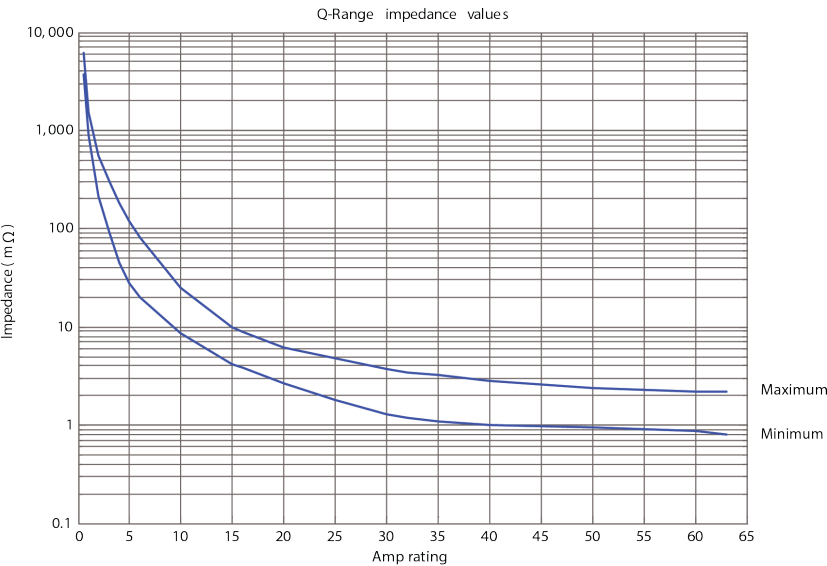


Time Delay Curves

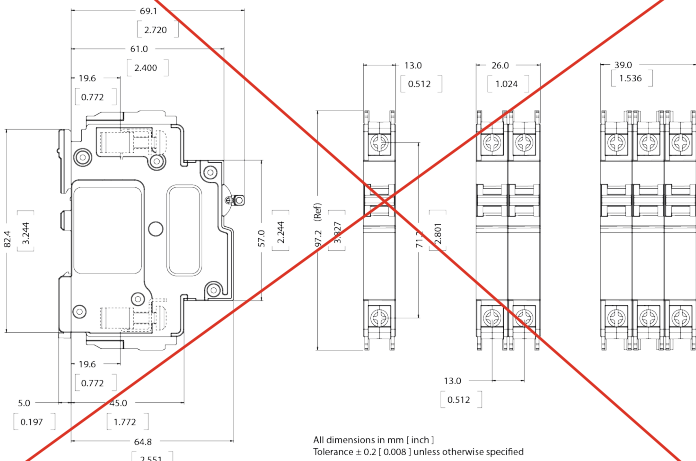


* The published time delay curves are generated at 30°C ambient temperature with the Circuit Breaker mounted in the up-right position. The “must hold”, “must trip” and “instantaneous trip” current values are not affected by temperature, although delay time for the other operating current values may have to be adjusted using the temperature compensation curve which is available on request.

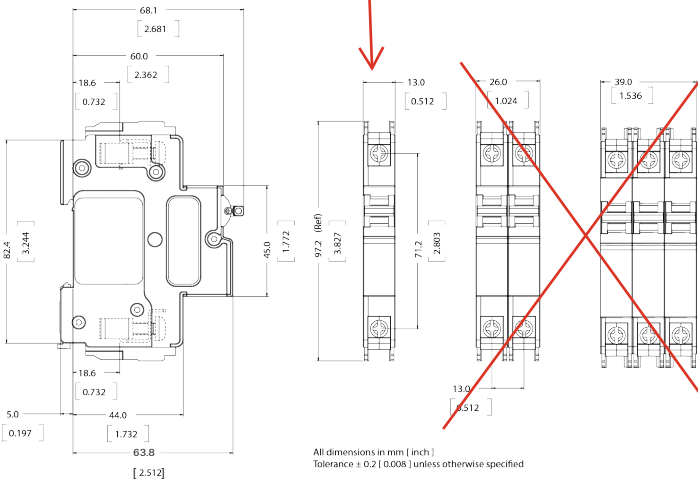
Internal Impedance vs Current Rating



Outline Dimensions: Dual mount



Outline Dimensions: DIN mount



Data Sheet 5

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
9	Mfg: Schneider Electric: Miniature Relay, 120VAC, 8 Blade Model Number: RXM2AB2F7 Tags/Service: CR4209, CR4210, CR4223, CR4225, CR4226, CR4306, CR4310, CR4316, CR4317 / Secondary Control Panel Specifications: • 120VAC operational voltage • 2 C/O contact configuration • With status LED and lockable test button • 10A continuous output current • 12A conventional enclosed thermal current • 12A rated operational current • 100000 cycles resistive electrical durability • Silver nickel contacts material
9	Mfg: Schneider Electric: Miniature Relay Socket, 8 Blade, DIN Model Number: RXZE2M114 Tags/Service: CR4209, CR4210, CR4223, CR4225, CR4226, CR4306, CR4310, CR4316, CR4317 / Secondary Control Panel Specifications: • 120VAC operational voltage • Compatible with RXM 2 C/O • 10A conventional free air thermal current

	• < 300 V UL rated operational voltage • Flat pin shape • < 1 Nm M3 tightening torque
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Product data sheet

MINIATURE RELAY, 120VAC, 8 BLADE

Specifications

TAGS: CR4209, CR4210, CR4223, CR4225, CR4226, CR4306
CR4310, CR4316, CR4317



miniature plug in relay, Harmony
Electromechanical Relays, 12A,
2CO, with LED, lockable test but to
n, 120V AC

RXM2AB2F7

Product availability: Stock - Normally stocked in distribution
facility

Main

Range of Product	Harmony Electromechanical Relays
Series name	RXM series
Product or Component Type	Plug-in relay
Relay Type	Miniature relay
Contacts type and composition	2 C/O
Status LED	With
Control Type	Lockable test button
[Uc] control circuit voltage	120 V AC 50/60 Hz
[Ithe] conventional enclosed thermal current	12 A
Continuous output current	10 A

Complementary

[Uimp] rated impulse withstand voltage	4 kV 1.2/50 µs
[Ie] rated operational current	12 A 28 V DC) NO-IEC 12 A 250 V AC) NO-IEC 6 A 28 V DC) NC-IEC 6 A 250 V AC) NC-IEC 12 A 28 V DC) UL 12 A 277 V AC) UL
Minimum switching capacity	170 mW 10 mA, 17 V
Electrical durability	100000 cycles resistive
Average coil consumption in VA	1.2 60 Hz
Rated operational voltage limits	96...132 V AC
[Ui] rated insulation voltage	250 V IEC 300 V CSA 300 V UL
Average consumption	1.2 VA 60 Hz
Maximum switching voltage	250 V IEC
Drop-out voltage threshold	>= 0.15 Uc
Load current	12 A 250 V AC 12 A 28 V DC
Operating time	20 ms
Maximum switching capacity	3000 VA/336 W
Average resistance	4430 Ohm 20 °C +/- 15 %

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Mechanical durability	10000000 cycles
Safety reliability data	B10d = 100000
Operating rate	<= 1200 cycles/hour under load <= 18000 cycles/hour no-load
Utilisation coefficient	20 %
CAD overall height	3.1 in (79 mm)
CAD overall depth	3.09 in (78.45 mm)
reset time	20 ms
Dielectric strength	1300 V AC between contacts with micro disconnection 2000 V AC between coil and contact with basic insulation 2000 V AC between poles with basic insulation
Compatibility code	RXM
Protection category	RT I
Pollution degree	3
Operating position	Any position
Test levels	Level A group mounting
Device presentation	Complete product
Contacts material	AgNi
Shape of pin	Flat (faston type)
Net Weight	0.082 lb(US) (0.037 kg)

Environment

Ambient air temperature for operation	-40...131 °F (-40...55 °C)
IP degree of protection	IP40 conforming to IEC 60529
Standards	CSA C22.2 No 14 IEC 61810-1 UL 508
Product Certifications	UL Lloyd's CE CSA GOST IECEE CB Scheme
Ambient Air Temperature for Storage	-40...185 °F (-40...85 °C)
Vibration resistance	3 gn +/- 1 mm 10...150 Hz)5 cycles in operation 5 gn +/- 1 mm 10...150 Hz)5 cycles not operating
Shock resistance	10 gnin operation 30 gnnot operating

Ordering and shipping details

Category	US10CP221127
Discount Schedule	0CP2
GTIN	3389119403504
Returnability	Yes
Country of origin	CN

Packing Units

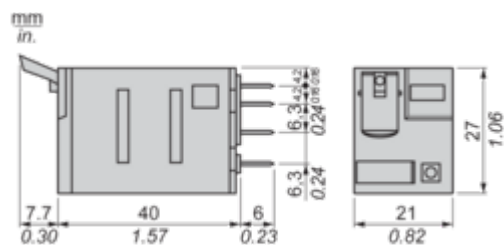
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	0.827 in (2.100 cm)
Package 1 Width	1.063 in (2.700 cm)
Package 1 Length	1.811 in (4.600 cm)
Package weight(Lbs)	1.305 oz (37.000 g)
Unit Type of Package 2	BB1
Number of Units in Package 2	10
Package 2 Height	1.181 in (3.000 cm)
Package 2 Width	4.016 in (10.200 cm)
Package 2 Length	4.921 in (12.500 cm)
Package 2 Weight	13.686 oz (388.000 g)
Unit Type of Package 3	S02
Number of Units in Package 3	240
Package 3 Height	5.906 in (15.000 cm)
Package 3 Width	11.811 in (30.000 cm)
Package 3 Length	15.748 in (40.000 cm)
Package 3 Weight	21.625 lb(US) (9.809 kg)

Contractual warranty

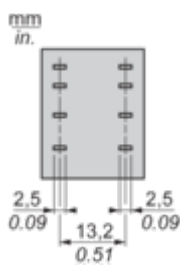
Warranty (in months)	18
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Dimensions Drawings

Dimensions



Pin Side View



Connections and Schema

Wiring Diagram

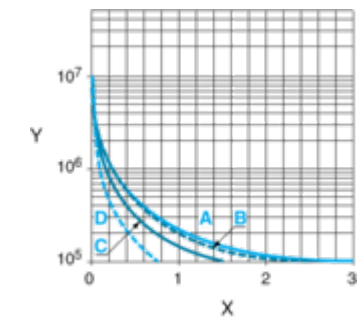


Symbols shown in blue correspond to Nema marking.

Performance Curves

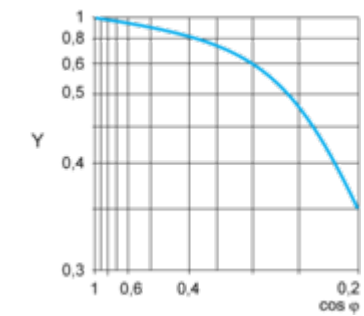
Electrical Durability of Contacts

Durability (inductive load) = durability (resistive load) x reduction coefficient.
Resistive AC load

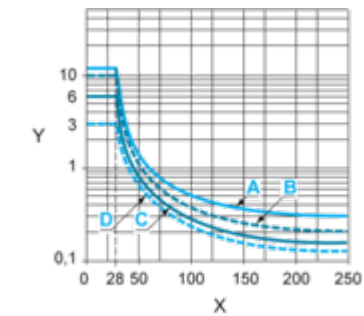


- X Switching capacity (kVA)
- Y Durability (Number of operating cycles)
- A RXM2AB...
- B RXM3AB...
- C RXM4AB...
- D RXM4GB...

Reduction coefficient for inductive AC load (depending on power factor cos φ)



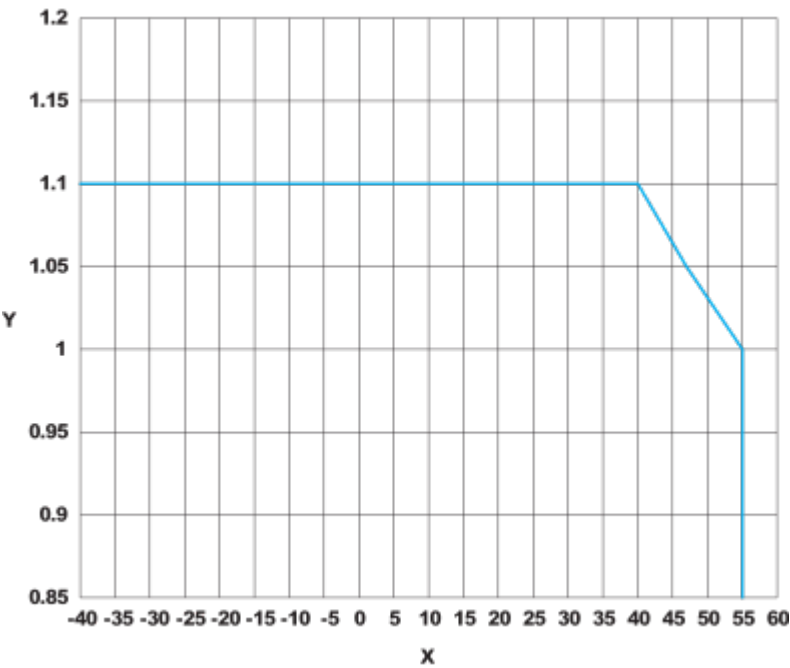
- Y Reduction coefficient (A)
- Maximum switching capacity on resistive DC load



- X Voltage DC
- Y Current DC
- A RXM2AB...
- B RXM3AB...
- C RXM4AB...
- D RXM4GB...

Note : These are typical curves, actual durability depends on load, environment, duty cycle, etc.
For inductive load, to increase relay life cycles, please add a proper load protection circuit (eg: RC protection/Varistor/free Wheeling diode -DC load only-).
For low level loads (below 10mA), we recommend to use RXM*GB series with bifurcated contacts relays instead.

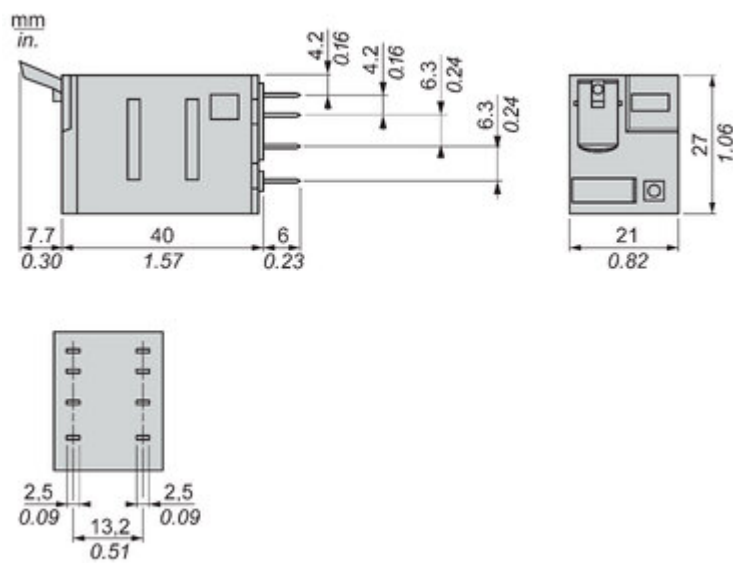
AC Coil Voltage and Operating Temperature under continuous duty



X : Operating temperature (°C)
Y : AC coil voltage (UC)

Technical Illustration

Dimensions



Product data sheet

MINIATURE RELAY SOCKET, 8 BLADE, DIN

Specifications

TAGS: CR4209, CR4210, CR4223, CR4225, CR4226, CR4306
CR4310, CR4316, CR4317



Harmony, plugin relay socket, mixed contact, 10 A, 250 V, screw clamp, for RXM2 or RXM4 relays

RXZE2M114

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony Electromechanical Relays Harmony Timer Relays
Product compatibility	Plug-in relay RXM 2 C/O Plug-in relay RXM 4 C/O Plug-in relay REXL 2 C/O Plug-in relay REXL 4 C/O Plug-in relay 782H 4 C/O
Device short name	RXZ
[Ith] conventional free air thermal current	10 A
Product or Component Type	Socket
Contacts type and composition	4 C/O

Complementary

[Ue] rated operational voltage	< 250 V IEC < 300 V UL
Connections - terminals	Screw clamp terminals, 1 x 0.34...1 x 1 mm² AWG 22...AWG 17) flexible with cable end Screw clamp terminals, 2 x 0.34...2 x 1 mm² AWG 22...AWG 17) flexible with cable end Screw clamp terminals, 1 x 0.5...1 x 1.5 mm² AWG 20...AWG 16) solid without cable end Screw clamp terminals, 2 x 0.5...2 x 1.5 mm² AWG 20...AWG 16) solid without cable end
Shape of pin	Flat
Tightening torque	< 1 N.m M3
Fixing mode	By screw (panel) Clip-on (35 mm symmetrical DIN rail)
Wire stripping length	0.3 in (8 mm)
Contact terminal arrangement	Mixed
[Uimp] rated impulse withstand voltage	4 kV
Overvoltage category	III
Pollution degree	3
Sale per indivisible quantity	10
Width	1.2 in (30 mm)
Net Weight	0.112 lb(US) (0.051 kg)

Environment

IP degree of protection	IP20 conforming to IEC 60529
-------------------------	------------------------------

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Dielectric strength	2500 V
Standards	IEC 61984
Product Certifications	UL CSA Lloyd's
Marking	CE
Ambient Air Temperature for Storage	-40...185 °F (-40...85 °C)
Ambient Air Temperature for Operation	-40...131 °F (-40...55 °C)

Ordering and shipping details

Category	US10CP221128
Discount Schedule	0CP2
GTIN	3389119404242
Returnability	Yes
Country of origin	CN

Packing Units

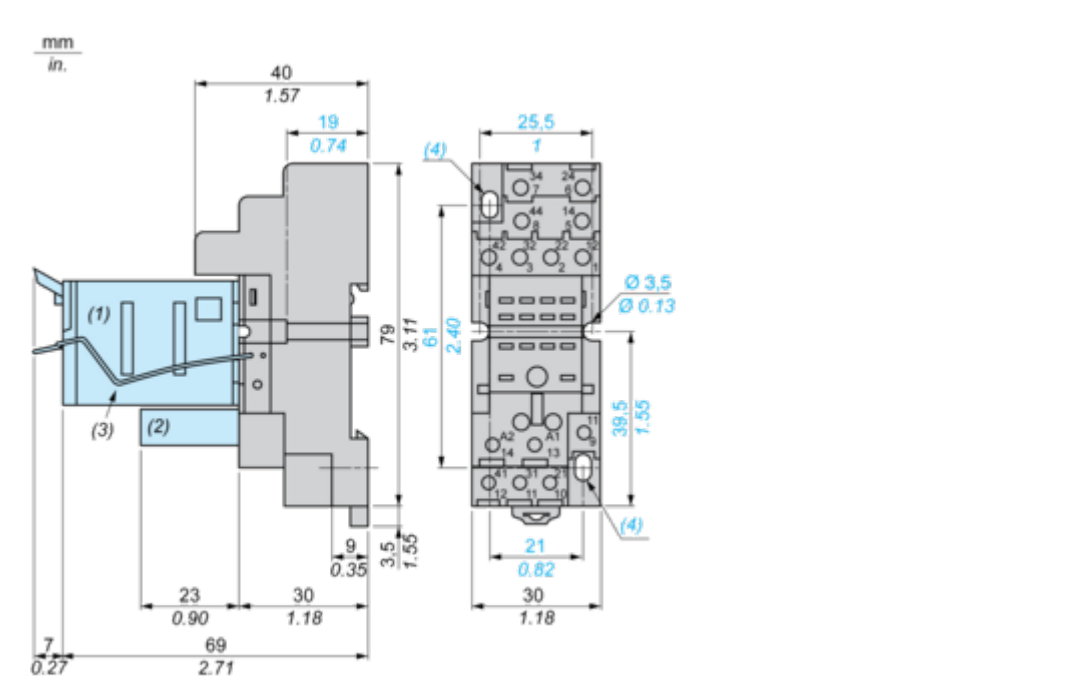
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.024 in (2.600 cm)
Package 1 Width	1.220 in (3.100 cm)
Package 1 Length	2.756 in (7.000 cm)
Package weight(Lbs)	1.975 oz (56.000 g)
Unit Type of Package 2	BB1
Number of Units in Package 2	10
Package 2 Height	2.992 in (7.600 cm)
Package 2 Width	3.504 in (8.900 cm)
Package 2 Length	6.102 in (15.500 cm)
Package 2 Weight	21.164 oz (600.000 g)
Unit Type of Package 3	S03
Number of Units in Package 3	200
Package 3 Height	11.811 in (30.000 cm)
Package 3 Width	11.811 in (30.000 cm)
Package 3 Length	15.748 in (40.000 cm)
Package 3 Weight	27.445 lb(US) (12.449 kg)

Contractual warranty

Warranty (in months)	18
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Dimensions Drawings

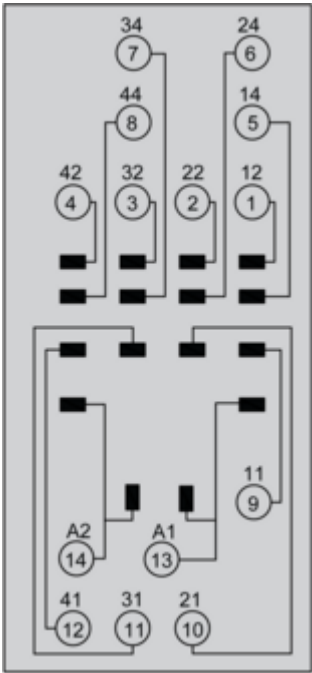
Dimensions



- (1) Relays
- (2) Protection module
- (3) Maintaining clamp
- (4) 2 elongated holes $\varnothing 3.5$ mm x 6.5 mm / $\varnothing 0.13$ in. x 0.25 in.

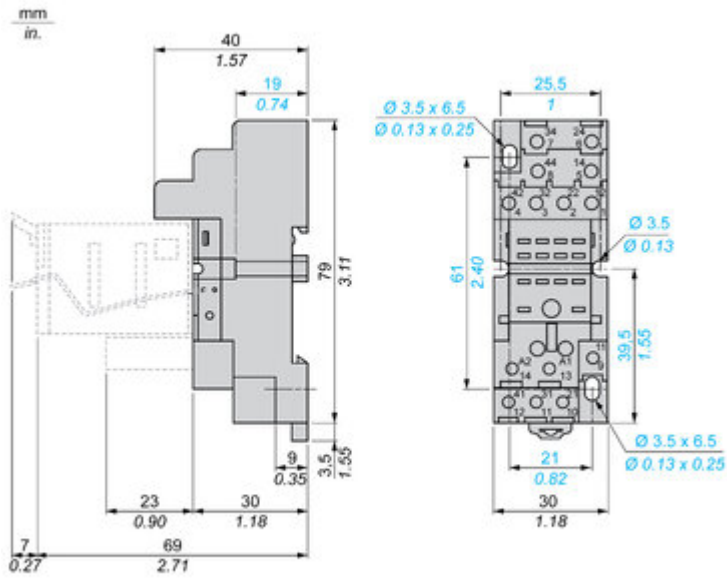
Connections and Schema

Wiring Diagram



Technical Illustration

Dimensions



Data Sheet 6

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Schneider Electric: Miniature Relay, 120VAC, 11 Blade Model Number: RXM3AB2F7 Tags/Service: CR4301 / Secondary Control Panel Specifications: • 120VAC operational voltage • 3 C/O contact configuration • With status LED and lockable test button • 6.7A continuous output current • 10A conventional enclosed thermal current • 10A rated operational current • 100000 cycles resistive electrical durability • Silver nickel contacts material
1	Mfg: Schneider Electric: Miniature Relay Socket, 11 Blade, DIN Model Number: RXZE2S111M Tags/Service: CR4301 / Secondary Control Panel Specifications: • 120VAC operational voltage • Compatible with RXM 3 C/O • 10A conventional free air thermal current • < 300 V UL rated operational voltage • Flat pin shape

	• < 1 Nm M3 tightening torque
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Product data sheet

MINIATURE RELAY, 120VAC, 11 BLADE

Specifications

TAGS: CR4301



miniature plug in relay, Harmony Electromechanical Relays, 10A, 3CO, with LED, lockable test but to n, 120V AC

RXM3AB2F7

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony Electromechanical Relays
Series name	RXM series
Product or Component Type	Plug-in relay
Relay Type	Miniature relay
Contacts type and composition	3 C/O
Status LED	With
Control Type	Lockable test button
[Uc] control circuit voltage	120 V AC 50/60 Hz
[Ithe] conventional enclosed thermal current	10 A
Continuous output current	6.7 A

Complementary

[Uimp] rated impulse withstand voltage	4 kV 1.2/50 µs
[Ie] rated operational current	10 A 28 V DC) NO-IEC 10 A 250 V AC) NO-IEC 5 A 28 V DC) NC-IEC 5 A 250 V AC) NC-IEC 10 A 30 V DC) UL 10 A 277 V AC) UL
Minimum switching capacity	170 mW 10 mA, 17 V
Electrical durability	100000 cycles resistive
Average coil consumption in VA	1.2 60 Hz
Rated operational voltage limits	96...132 V AC
[Ui] rated insulation voltage	250 V IEC 300 V CSA 300 V UL
Average consumption	1.2 VA 60 Hz
Maximum switching voltage	250 V IEC
Drop-out voltage threshold	>= 0.15 Uc
Load current	10 A 250 V AC 10 A 28 V DC
Operating time	20 ms
Maximum switching capacity	2500 VA/280 W
Average resistance	4430 Ohm 20 °C +/- 15 %

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Mechanical durability	10000000 cycles
Safety reliability data	B10d = 100000
Operating rate	<= 1200 cycles/hour under load <= 18000 cycles/hour no-load
Utilisation coefficient	20 %
CAD overall height	3.1 in (79 mm)
CAD overall depth	3.09 in (78.45 mm)
reset time	20 ms
Dielectric strength	1300 V AC between contacts with micro disconnection 2000 V AC between coil and contact 2000 V AC between poles
Compatibility code	RXM
Protection category	RT I
Pollution degree	2
Operating position	Any position
Test levels	Level A group mounting
Device presentation	Complete product
Contacts material	AgNi
Shape of pin	Flat (faston type)
Net Weight	0.082 lb(US) (0.037 kg)

Environment

Ambient air temperature for operation	-40...131 °F (-40...55 °C)
IP degree of protection	IP40 conforming to IEC 60529
Standards	UL 508 IEC 61810-1 CSA C22.2 No 14
Product Certifications	UL Lloyd's CE CSA EAC IECEE CB Scheme
Ambient Air Temperature for Storage	-40...185 °F (-40...85 °C)
Vibration resistance	3 gn +/- 1 mm 10...150 Hz)5 cycles in operation 5 gn +/- 1 mm 10...150 Hz)5 cycles not operating
Shock resistance	10 gnin operation 30 gnnot operating

Ordering and shipping details

Category	US10CP221127
Discount Schedule	0CP2
GTIN	3389119403665
Returnability	Yes
Country of origin	CN

Packing Units

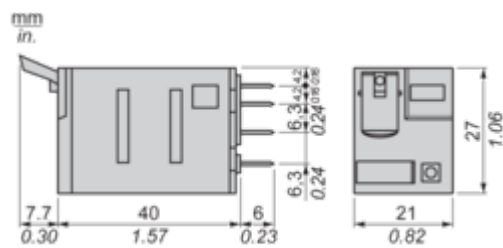
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	0.984 in (2.500 cm)
Package 1 Width	1.181 in (3.000 cm)
Package 1 Length	1.772 in (4.500 cm)
Package weight(Lbs)	1.305 oz (37.000 g)
Unit Type of Package 2	BB1
Number of Units in Package 2	10
Package 2 Height	1.181 in (3.000 cm)
Package 2 Width	4.134 in (10.500 cm)
Package 2 Length	4.921 in (12.500 cm)
Package 2 Weight	13.968 oz (396.000 g)
Unit Type of Package 3	S02
Number of Units in Package 3	240
Package 3 Height	5.906 in (15.000 cm)
Package 3 Width	11.811 in (30.000 cm)
Package 3 Length	15.748 in (40.000 cm)
Package 3 Weight	21.497 lb(US) (9.751 kg)

Contractual warranty

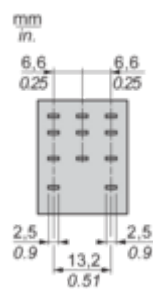
Warranty (in months)	18
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Dimensions Drawings

Dimensions



Pin Side View



Connections and Schema

Wiring Diagram

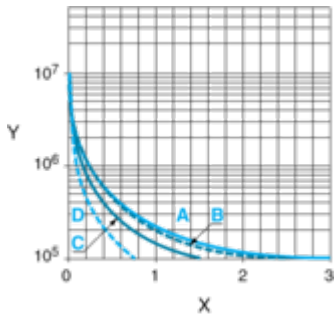


Symbols shown in blue correspond to Nema marking.

Performance Curves

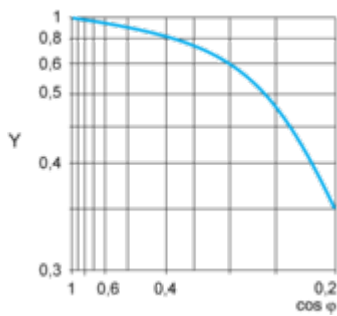
Electrical Durability of Contacts

Durability (inductive load) = durability (resistive load) x reduction coefficient.
Resistive AC load

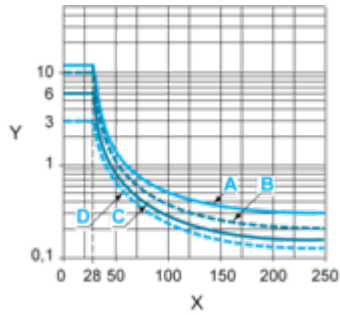


- X Switching capacity (kVA)
- Y Durability (Number of operating cycles)
- A RXM2AB...
- B RXM3AB...
- C RXM4AB...
- D RXM4GB...

Reduction coefficient for inductive AC load (depending on power factor cos φ)



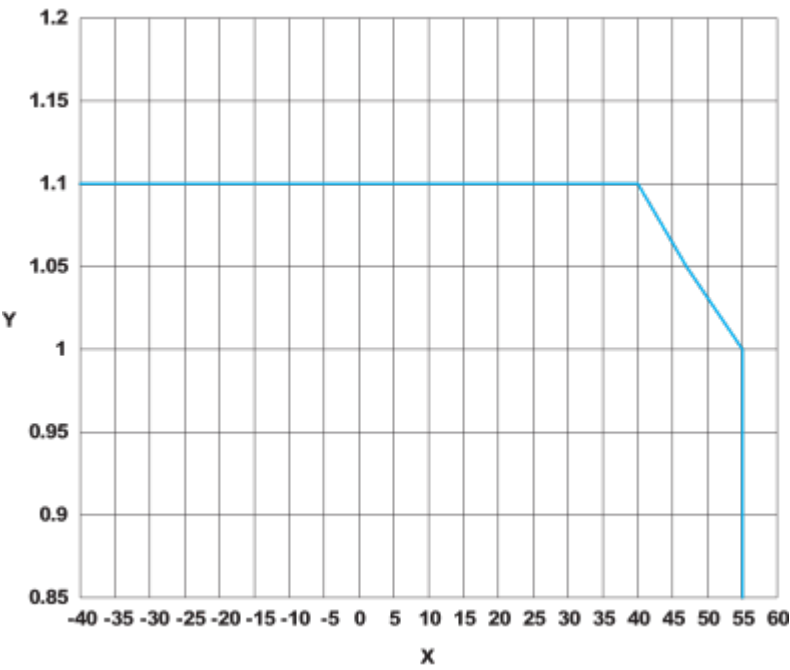
- Y Reduction coefficient (A)
- Maximum switching capacity on resistive DC load



- X Voltage DC
- Y Current DC
- A RXM2AB...
- B RXM3AB...
- C RXM4AB...
- D RXM4GB...

Note : These are typical curves, actual durability depends on load, environment, duty cycle, etc.
For inductive load, to increase relay life cycles, please add a proper load protection circuit (eg: RC protection/Varistor/free Wheeling diode -DC load only-).
For low level loads (below 10mA), we recommend to use RXM*GB series with bifurcated contacts relays instead.

AC Coil Voltage and Operating Temperature under continuous duty

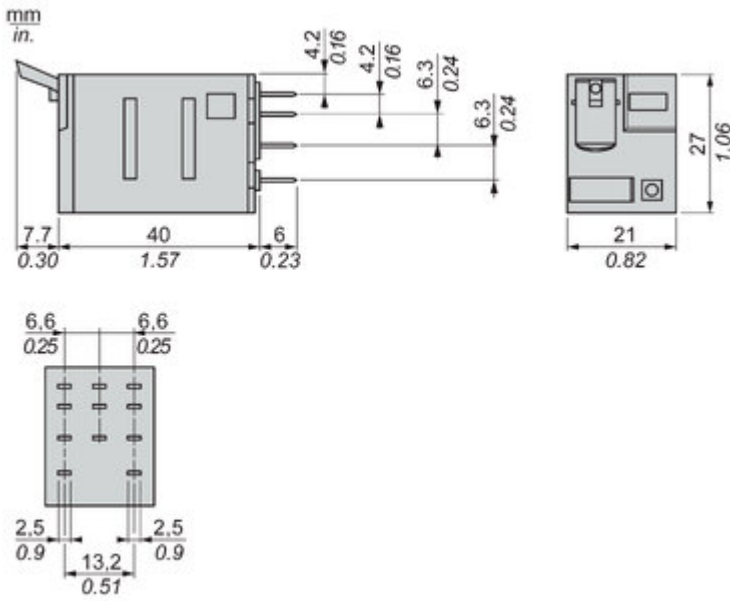


X : Operating temperature (°C)

Y : AC coil voltage (UC)

Technical Illustration

Dimensions



TAGS: CR4301



socket, Harmony Electromechanical Relays, for RXM3 relays, screw connec to rs, separate contact

RXZE2S111M

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony Electromechanical Relays
Product compatibility	Plug-in relay RXM 3 C/O
Device short name	RXZ
[Ith] conventional free air thermal current	10 A
Product or Component Type	Socket
Contacts type and composition	3 C/O

Complementary

[Ue] rated operational voltage	< 250 V IEC < 300 V UL
Connections - terminals	Screw connectors, 1 x 0.34...1 x 2.5 mm² AWG 22...AWG 14) flexible with cable end Screw connectors, 2 x 0.34...2 x 1 mm² AWG 22...AWG 17) flexible with cable end Screw connectors, 1 x 0.5...1 x 2.5 mm² AWG 20...AWG 14) solid without cable end Screw connectors, 2 x 0.5...2 x 1.5 mm² AWG 20...AWG 16) solid without cable end
Shape of pin	Flat
Tightening torque	< 1 N.m M3
Fixing mode	By screw (panel) Clip-on (35 mm symmetrical DIN rail)
Wire stripping length	0.3 in (8 mm)
Contact terminal arrangement	Separate
[Uimp] rated impulse withstand voltage	4 kV
Overvoltage category	III
Pollution degree	3
Sale per indivisible quantity	10
Width	1.06 in (27 mm)
Net Weight	0.128 lb(US) (0.058 kg)

Environment

IP degree of protection	IP20 conforming to IEC 60529
Dielectric strength	2500 V
Standards	IEC 61984
Product Certifications	UL Lloyd's CSA
Marking	CE

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Ambient Air Temperature for Storage	-40...185 °F (-40...85 °C)
Ambient Air Temperature for Operation	-40...131 °F (-40...55 °C)

Ordering and shipping details

Category	US10CP221128
Discount Schedule	0CP2
GTIN	3389119404273
Returnability	Yes
Country of origin	CN

Packing Units

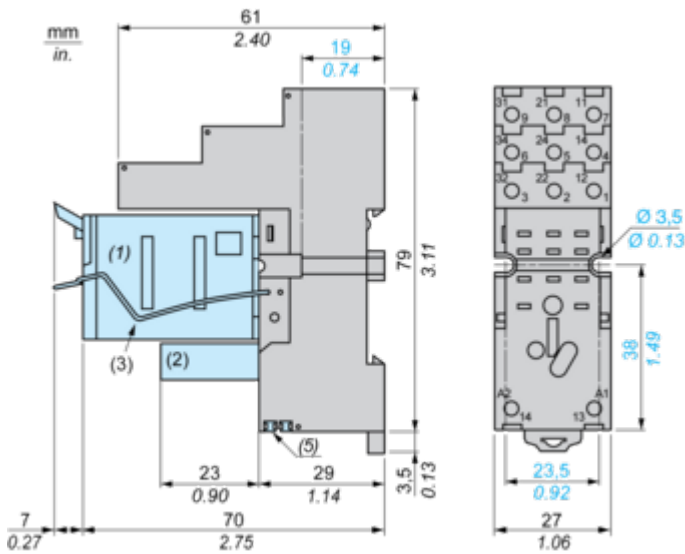
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	2.402 in (6.100 cm)
Package 1 Width	1.063 in (2.700 cm)
Package 1 Length	2.953 in (7.500 cm)
Package weight(Lbs)	2.116 oz (60.000 g)
Unit Type of Package 2	BB1
Number of Units in Package 2	10
Package 2 Height	2.598 in (6.600 cm)
Package 2 Width	5.354 in (13.600 cm)
Package 2 Length	4.331 in (11.000 cm)
Package 2 Weight	22.787 oz (646.000 g)
Unit Type of Package 3	S03
Number of Units in Package 3	200
Package 3 Height	11.811 in (30.000 cm)
Package 3 Width	11.811 in (30.000 cm)
Package 3 Length	15.748 in (40.000 cm)
Package 3 Weight	29.568 lb(US) (13.412 kg)

Contractual warranty

Warranty (in months)	18
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Dimensions Drawings

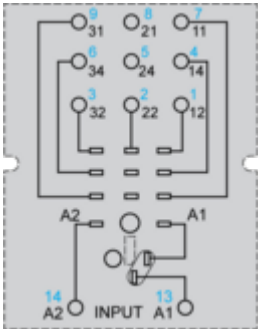
Dimensions



- (1) Relays
- (2) Protection module
- (3) Maintaining clamp
- (4) 2 elongated holes Ø 3.5 mm x 6.5 mm / Ø 0.13 in. x 0.25 in.
- (5) 2 bus jumpers

Connections and Schema

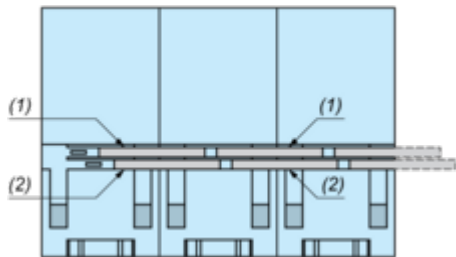
Wiring Diagram



Symbols shown in blue correspond to Nema marking.

Bus Jumpers Mounting on Sockets with Separate Contacts

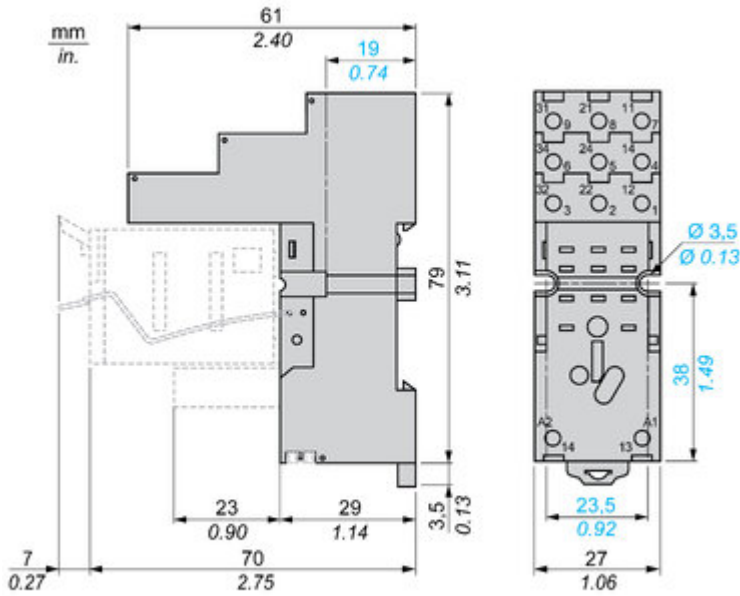
Example of RXZS2 bus jumper mounting on sockets (view from below)



- (1) 2 bus jumpers (polarity A2)
- (2) 2 bus jumpers (polarity A1)

Technical Illustration

Dimensions



Data Sheet 7

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Square D: Timing Relay, 0-10 min, 120VAC, 8 Pin Octal, DPDT Model Number: 9050JCK16V20 Tags/Service: TR4224 / Secondary Control Panel Specifications: • 120VAC operational voltage • 0.1...10 min timing range • 2VA power consumption • 10A operational current rating • DPDT, 2 NO + 2 NC contaction composition • Silver nickel contact material
1	Mfg: Square D: Timing Relay Socket, 8 Pin, DIN Model Number: 8501NR51 Tags/Service: TR4224 / Secondary Control Panel Specifications: • 120VAC operational voltage • 15A rated current • Single tiered screw clamp terminal electrical connection

Product data sheet

TIMING RELAY, ON DELAY

Specifications

TAGS: TR4224



Timer, Harmony 9050 JCK, on delay, adjustable time, 0.1 to 10 minutes, 10A, 240V AC, 120V AC/110V DC

9050JCK16V20

Product availability: Stock - Normally stocked in distribution facility

Main

product range	9050JCK
product type	Timer
timing mode	On-delay
timing range	0.1...10 min
supply voltage	120 V AC 110 V DC

Complementary

power consumption	2 VA
current rating	10 A
switch function	DPDT
contact composition	2 NO + 2 NC
contact material	Silver nickel
maximum switching voltage	240 V AC
mounting type	Socket
number of pins	8
shape of pins	Tubular
for use with	8501NR51 8501NR52

Environment

certifications	CE UL Listed UL Recognized
ambient temperature range - operational	-18...150 °F (-28...66 °C)
Height	2.4 in (61 mm)
Width	1.7 in (43 mm)
Depth	4.09 in (104 mm)

Ordering and shipping details

Category	US10CP221651
Discount Schedule	0CP2
GTIN	785901032595

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Returnability	Yes
Country of origin	US

Packing Units

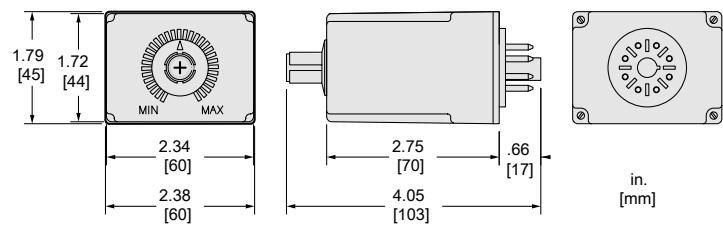
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	2.00 in (5.08 cm)
Package 1 Width	2.600 in (6.604 cm)
Package 1 Length	4.400 in (11.176 cm)
Package weight(Lbs)	5.280 oz (149.686 g)
Unit Type of Package 2	CAR
Number of Units in Package 2	20
Package 2 Height	4.50 in (11.43 cm)
Package 2 Width	10.0 in (25.4 cm)
Package 2 Length	11.00 in (27.94 cm)
Package 2 Weight	5.501 lb(US) (2.495 kg)

Contractual warranty

Warranty (in months)	18
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Technical Illustration

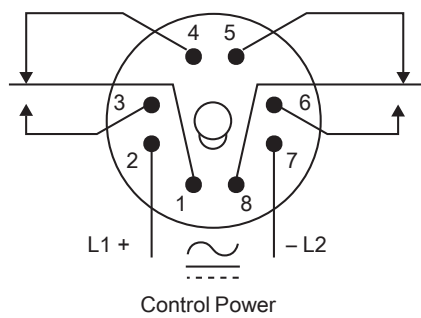
Dimensions



All dimensions are approximate. Also see technical documentation.

Technical Illustration

Wiring diagram



- JCK11 – JCK19
- JCK31 – JCK39
- JCK51 – JCK59
- JCK60

Product data sheet

TIMING RELAY SOCKET

Specifications

TAGS: TR4224

SQUARE D



Plug in relay, Type N, relay socket, 8 tubular pin, single tier, for 8501KP relays and 9050JCK timers

8501NR51

Product availability: Stock - Normally stocked in distribution facility

Main

[In] rated current	15 A 300 V AC 5 A 600 V AC
--------------------	-------------------------------

Complementary

mounting type	DIN rail Panel
electrical connection	Single tiered screw clamp terminals
package quantity	1
Width	1.6 in (40 mm)
Height	2.1 in (54 mm)
Depth	1.0 in (24.6 mm)

Ordering and shipping details

Category	US10CP221142
Discount Schedule	0CP2
GTIN	785901583592
Returnability	Yes
Country of origin	CN

Packing Units

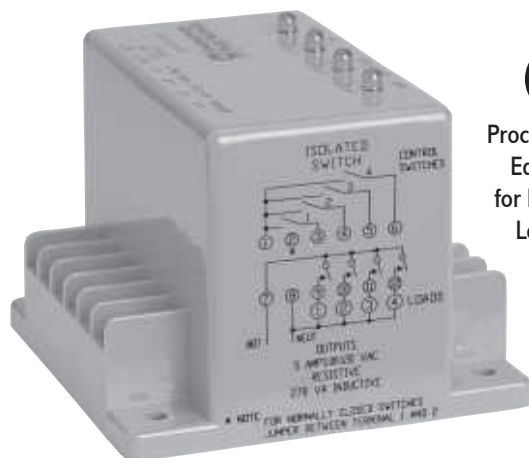
Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.000 in (2.540 cm)
Package 1 Width	1.700 in (4.318 cm)
Package 1 Length	2.200 in (5.588 cm)
Package weight(Lbs)	1.600 oz (45.359 g)
Unit Type of Package 2	CAR
Number of Units in Package 2	600
Package 2 Height	14.200 in (36.068 cm)
Package 2 Width	11.200 in (28.448 cm)
Package 2 Length	17.800 in (45.212 cm)
Package 2 Weight	64.0002 lb(US) (29.030 kg)

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Data Sheet 8

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
2	Mfg: Diversified: Isolated Switch, 2 Channel Model Number: ISO-120-AAE Tags/Service: ISR4206, ISR4220 / Secondary Control Panel Specifications: • 120VAC control voltage, 2 channel • 2.5VA power consumption • 20 million operations mechanical life expectancy • 50,000 operations at rated load electrical life expectancy • -4° to 131° F operational temperature rating • -40° to 185° F storage temperature rating • Style “E” Lexan® surface mounted




 Process Control
 Equipment
 for Hazardous
 Locations
 7M26
 UL913

Multiple Channel Isolated Switch

SPECIFICATIONS

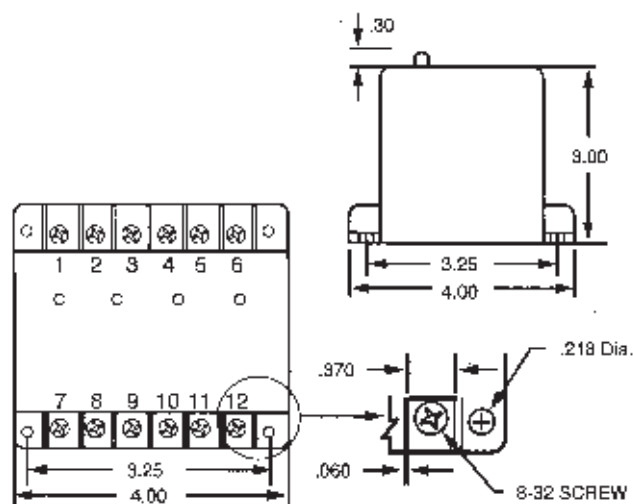
CONTROL VOLTAGE	24 or 120 VAC $\pm 10\%$, 50/60 Hz	
CONTROL SWITCH	Open Circuit Voltage	6.2 VDC
	Short Circuit Current	10 μ Amps
RESPONSE TIMES	Operate	6 mSEC (Approx.)
	Release	2.5 mSEC (Approx.)
POWER REQUIRED	2.5 VA	
DUTY CYCLE	Continuous	
CONTACT RATING	SPST-N.O., 5 amps per channel @ 24 or 120 VAC, Resistive; 278 VA, Inductive	
ISOLATION	2500 Volts, Input to Output	
LIFE EXPECTANCY	Mechanical	20 Million Operations
	Electrical	50,000 Operations @ Rated Load
INDICATORS	On When Corresponding Output is On	
TEMPERATURE RATING	Operate	-4° to 131°F (-20° to +55°C)
	Storage	-40° to 185°F (-40° to +85°C)
ENCLOSURE	Style "E" Lexan® Surface Mounted	
TERMINATIONS	(12) #8-32 Screw terminals	
WEIGHT	8 oz.	

OPERATION

The **ISO/ISL Series multiple channel** devices are used to provide a safe and reliable means of controlling loads from hazardous locations without releasing sufficient energy under normal or abnormal conditions to cause ignition of a flammable or combustible atmospheric mixture while in its most easily ignited concentration. An isolated output turns on when the corresponding control switch input from the hazardous location is activated. When using normally closed control switch inputs, a jumper should be installed between terminals 1 and 2. Normally open control switch inputs do not require the optional jumper. When the

non-latching ISO Series control switch input is activated, its corresponding output turns on. When the control switch input is deactivated, its output turns off. ~~When the **latching ISL Series** control switch input 2, 3 or 4 is activated, its corresponding output turns on. When control switch 2, 3 or 4 is deactivated, its corresponding output remains latched on as long as control switch input 1 is activated; otherwise it turns off immediately.~~ Control switch input 1 also controls output 1 just as in the non-latching ISO Series.

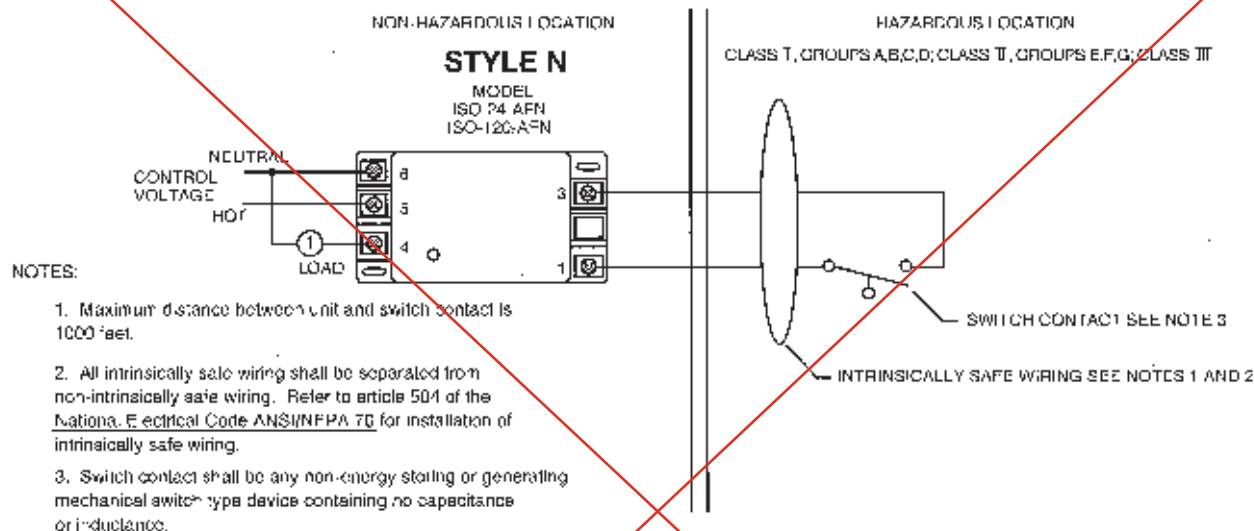
DIMENSIONS (INCHES)



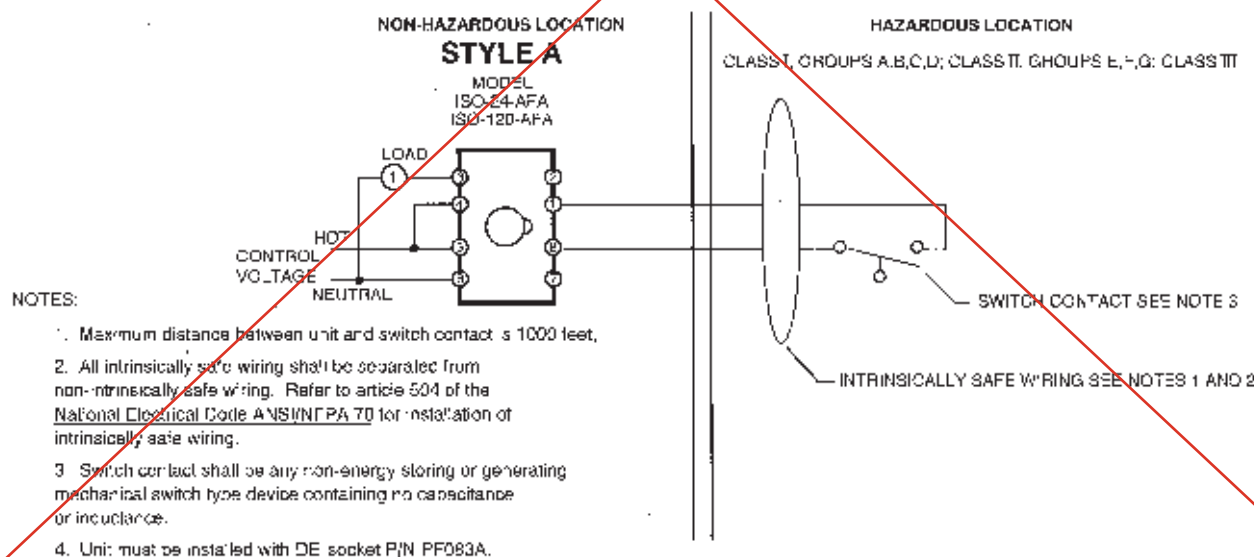
ORDERING INFORMATION

MODEL NUMBER	CONTROL VOLTAGE	CHANNELS
ISL-24-AAE	24 VAC	2
ISL-24-ABE	24 VAC	3
ISL-24-ACE	24 VAC	4
ISL-120-AAE	120 VAC	2
ISL-120-ABE	120 VAC	3
ISL-120-ACE	120 VAC	4
ISO-24-AAE	24 VAC	2
ISO-24-ABE	24 VAC	3
ISO-24-ACE	24 VAC	4
ISO-120-AAE	120 VAC	2
ISO-120-ABE	120 VAC	3
ISO-120-ACE	120 VAC	4

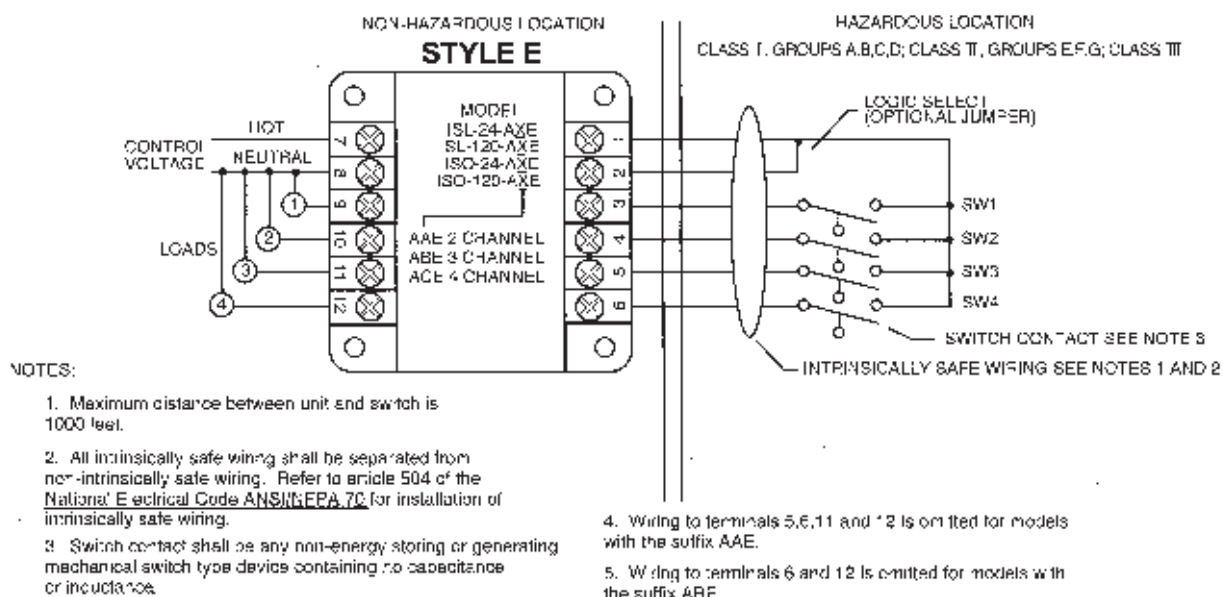
CONTROL DRAWING 191



CONTROL DRAWING 192



CONTROL DRAWING 193



Data Sheet 9

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Eaton: Alternating Relay Model Number: D851LA Tags/Service: AR4315 / Secondary Control Panel Specifications: • SPDT contact • With ALTERNATE, LOAD A, LOAD B selector switch • 120VAC control voltage • Less than 3VA load • 10,000,000 operations mechanical life expectancy • 100,000 operations electrical life • -20° to 150° F operating temperature range • 2 indicator LEDs for LOAD A and LOAD B
1	Mfg: Eaton: Alternating Relay Socket Model Number: D3PA2 Tags/Service: AR4315 / Secondary Control Panel Specifications: • General-purpose relay socket • D3 series socket • DIN rail mounted socket • 10/15A nominal rated operational current • Screw clamping wiring connection

TAGS: AR4315

Product Selection

D85 Series—Alternating Relays ^①

3

Output Contacts	Control Voltage	Socket	Catalog Number
SPDT	12 Vac	8-pin	D851NR
SPDT	24 Vac	8-pin	D851NT
SPDT	120 Vac	8-pin	D851NA
SPDT	240 Vac	8-pin	D851NB
SPDT w/selector switch	12 Vac	8-pin	D851LR
SPDT w/selector switch	24 Vac	8-pin	D851LT
SPDT w/selector switch	120 Vac	8-pin	D851LA
SPDT w/selector switch	240 Vac	8-pin	D851LB
DPDT	12 Vac	11-pin	D852NR
DPDT	24 Vac	11-pin	D852NT
DPDT	120 Vac	11-pin	D852NA
DPDT	240 Vac	11-pin	D852NB
DPDT w/selector switch	12 Vac	11-pin	D852LR
DPDT w/selector switch	24 Vac	11-pin	D852LT
DPDT w/selector switch	120 Vac	11-pin	D852LA
DPDT w/selector switch	240 Vac	11-pin	D852LB
DPDT cross-wired	12 Vac	8-pin	D852XNR
DPDT cross-wired	24 Vac	8-pin	D852XNT
DPDT cross-wired	120 Vac	8-pin	D852XNA
DPDT cross-wired	240 Vac	8-pin	D852XNB
DPDT cross-wired w/selector switch	12 Vac	8-pin	D852XLR
DPDT cross-wired w/selector switch	24 Vac	8-pin	D852XLT
DPDT cross-wired w/selector switch	120 Vac	8-pin	D852XLA
DPDT cross-wired w/selector switch	240 Vac	8-pin	D852XLB

Accessories

D85 Series—Alternating Relays

Description	Standard Pack	Catalog Number
8-pin socket	10	D3PA2
11-pin socket	10	D3PA3-A2
Hold-down spring	10	D65CHDS

Note

① Contact Eaton for relays for 3-switch applications (Lead-Lag-Stop).

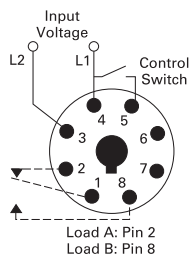
Technical Data and Specifications

D85 Series—Alternating Relays

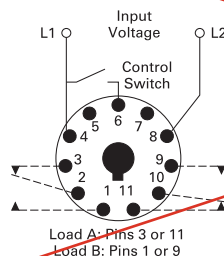
Description	Specification
Voltage tolerance	+10%/-15% of control voltage at 50/60 Hz
Load (burden)	Less than 3 VA
Output contacts	10 A resistive at 240 Vac / 30 Vdc, 1/2 hp at 120/240 Vac (NO), 1/3 hp at 120/240 Vac (NC)
Mechanical life	10,000,000 operations
Electrical life	100,000 operations
Temperature	-20 °F to +150 °F (-28 ° to 65 °C)
Transient protection	10,000 volts for 20 microseconds
Indicator LEDs	2 LEDs marked LOAD A and LOAD B
Optional selector switch settings	ALTERNATE, LOCK LOAD A, LOCK LOAD B

Wiring Diagrams

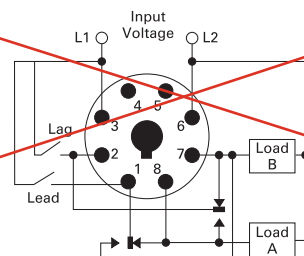
D851 Series Relays, SPDT



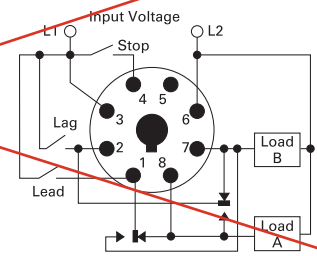
D852 Series Relays, DPDT



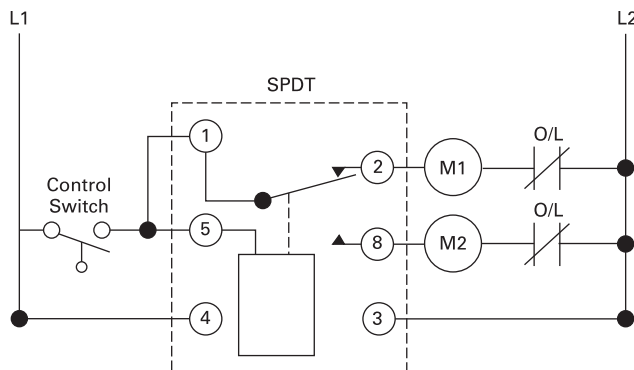
D852X Series Relays, DPDT Cross-Wired



D853 Series Relays, Three-Switch Applications



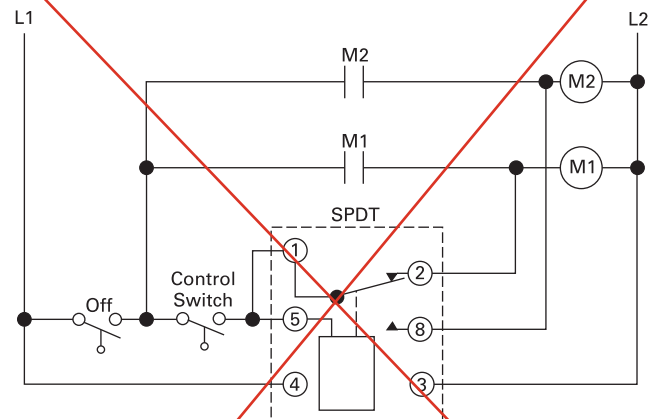
Typical Installations for SPDT and DPDT Alternating Relays, Standard Installation



In the OFF state (standard installation), the control switch is open, the alternating relay is in the LOAD A position, and both loads (M1 and M2) are off. When the control switch closes, it energizes the first load (M1). The red LED marked "LOAD A" glows. As long as the control switch remains closed, M1 remains energized.

When the control switch opens, the first load (M1) is turned off and the alternating relay toggles to the LOAD B position. When the control switch closes again, it energizes the second load (M2). The red LED marked "LOAD B" glows.

Typical Installations for SPDT and DPDT Alternating Relays, Anti-Bounce Installation

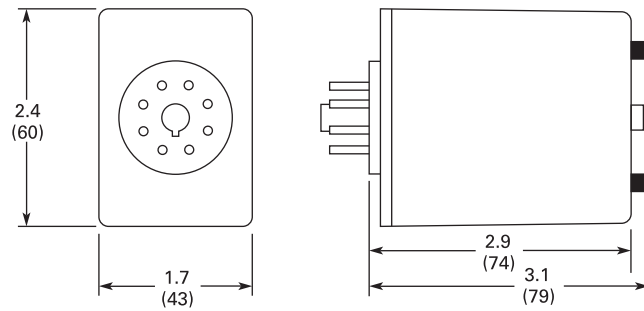


When the control switch opens, the second load (M2) is turned off, the alternating relay toggles back to the LOAD A position, and the process can be repeated again. On relays with DPDT contacts, two pilot lights can be used for remote indication of LOAD A or LOAD B status.

To eliminate any bounce condition of the control switch, the addition of a second switch (OFF) along with two auxiliary contacts is recommended as shown in the Anti-Bounce Installation.

Dimensions

Approximate Dimensions in Inches (mm)

D85 Series—Alternating Relays

Specifications

ALTERNATING RELAY SOCKET

TAGS: AR4315

Eaton D3PA2



D3 Series Socket, Used with D3PR2 and D3PF2 Relays, TRNP Timing Relays, TMR5 and TRM6 Time Delay Relays, DIN rail/panel mount, Screw clamping wire connection, QTY: 10, 8 pins

General specifications

PRODUCT NAME	Eaton General-purpose relay
CATALOG NUMBER	D3PA2
UPC	782114159767
PRODUCT LENGTH/DEPTH	2 in
PRODUCT HEIGHT	0.8 in
PRODUCT WIDTH	1.6 in
PRODUCT WEIGHT	0.07 lb
WARRANTY	Eaton Selling Policy 25-000, one (1) year from the date of installation of the Product or eighteen (18) months from the date of shipment of the Product, whichever occurs first.
COMPLIANCES	RoHS Compliant CE Marked
CERTIFICATIONS	CSA Certified UL Listed cRUus
PRODUCT TYPE	General-purpose relay

Product specifications

TYPE	D3 Series Socket
MOUNTING	DIN rail, screw
RATED OPERATIONAL CURRENT	10/15A nominal
USED WITH	D3PR2 and D3PF2 relays, TMR5 and TRM6 time delay relays, and TRNP timing relays
NOMINAL VOLTAGE	300/600 V
QUANTITY	10
SERIES	D3
WIRING CONNECTION	Screw clamping

Resources

CATALOGS	Motor Control and Protection, Motor Protection and Monitoring, Volume 5, Tab 5
SPECIFICATIONS AND DATASHEETS	Eaton Specification Sheet - D3PA2

PROJECT NAME:

PROJECT NUMBER:

PREPARED BY:

DATE:

Data Sheet 10

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: SCE: Enclosure Heater Model Number: SCE-TSH25 Tags/Service: HTR4202 / Secondary Control Panel Specifications: • DIN rail clip for 35mm DIN rail • Designed for protection from low temperatures, condensation, and corrosion • Touch safe with 2 screw terminal for standard AWG 14 wire • 120VAC operational voltage
1	Mfg: Finder: Enclosure Heater Thermostat, NC Model Number: 7T.81.0.000.2403 Tags/Service: THERMOSTAT / Secondary Control Panel Specifications: • -45...80° C ambient temperature range • 1 NC contact configuration • 10A rated current for contacts • 0...60° C temperature control range



ENCLOSURE HEATER

Saginaw Control and Engineering
 95 Midland Road Saginaw, MI 48638-5770
 (800) 234-6871 - Fax: (989) 799-4524
 SCE@SaginawControl.com

SCE-TSH25

TAGS: HTR4202

Product Specifications:



Part Number: SCE-TSH25
Description: Heater - 25W
Height: 4.92"
Width: 1.61"
Depth: 1.61"
Price Code: P1
List Price: \$111.42
Catalog Page: 402
Est. Ship Weight: 0.35 lbs
Max. Current : 2.5
Model Number: 501025

Application

PTC (Positive Temperature Coefficient)
 Din rail clip design for 35mm Din Rail. Designed for protection from low temperatures, condensation and corrosion. Touch Safe with 2 screw terminal for standard AWG 14 wire. 110-240V AC.

For low temperatures management use with thermostat

For condensate management no thermostat is necessary - continuous heat is most effective see installaton Instruction for sizing.

Industry Standards - (IS24)

✱ UL Component Recognized

Notes

UL File #E358386

Optional Accessories

~~SCE-TEMD Thermostat (Dual)~~
~~SCE-TEMNC Thermostat (Normally Closed)~~

Similar Part Numbers

~~SCE-TSH100Heater - 100W~~
~~SCE-TSH150Heater - 150W~~
~~SCE-TSH50Heater - 50W~~
~~SCE-TSH75Heater - 75W~~

Installation Information

✱ Touch Safe Heaters

7T8100002403

Panel Thermo-Hygrostat and Thermostats

TAGS: THERMOSTAT



Technical description:

Panel Thermo-Hygrostat and Thermostats, Temperature control, 1 pole, Monitoring function: Temperature control, NC, Control function: 0...+60 °C

This image is for illustrative purposes only and therefore may not represent the exact product code. Colours may vary.

[BROWSE PRODUCT PAGE](#)

Technical Features:

Ambient temperature range:	-45...80 °C
Commercial Description:	Panel thermostat, compact size (17.5 mm wide), controls the ON/OFF operation of a panel heater. Available in version 7T.81.0.000.230x for controlling ventilation.
Contact configuration:	1 NC
Dimensions:	17.5x84x43.5 mm
Function:	Monitoring function: Temperature control
Rated current (contacts):	10
Temperature range:	0 ... 60 °C
Certifications and quality approvals (according to type):	    

The information, technical details, data and features contained in this document are subject to change or modification at any time.
It is for information purposes only and does not form the basis of a contract.

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[Contact our sales](#) [Find a dealer](#)

Data Sheet 11

Customer:	Johnston Construction
Reference:	Charles County Pomonkey Pump Station Improvements
Date:	03/05/2026
<u>Qty</u>	<u>Description</u>
A/R	Mfg: Phoenix Contact: Terminal Block Model Number: 3044076
A/R	Mfg: Phoenix Contact: Grounding Terminal Block Model Number: 3044092
A/R	Mfg: Phoenix Contact: Terminal End Barrier Model Number: 3047028
A/R	Mfg: Phoenix Contact: Terminal Anchor Model Number: 0800886 Tags / Service: TB-AC, TB-ISB / Secondary Control Panel Specifications: • Feed through and grounding terminals • Screw clamps • End plates and anchors • DIN rail mount

UT 2,5

Order No.: 3044076



<http://eshop.phoenixcontact.de/phoenix/treeViewClick.do?UID=3044076>

Universal terminal block, Type of connection: Screw connection,
Screw connection, Cross section: 0.14 mm² - 4 mm², AWG 26 - 12,
Width: 5.2 mm, Color: gray, Mounting type: NS 35/7,5, NS 35/15



Commercial data	
EAN	4017918960377
Pack	50 pcs.
Customs tariff	85369010
Weight/Piece	0.008227 KG
Catalog page information	Page 26 (CL-2009)

Product notes

WEEE/RoHS-compliant since:
01/01/2003



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Technical data	
General	
Number of levels	1
Number of connections	2
Color	gray

Insulating material	PA
Inflammability class acc. to UL 94	V0

Dimensions

Width	5.2 mm
Length	47.7 mm
Height NS 35/7,5	47.5 mm
Height NS 35/15	55 mm

Technical data

Maximum load current	32 A (with 4 mm ² conductor cross section)
Rated surge voltage	8 kV
Pollution degree	3
Surge voltage category	III
Insulating material group	I
Connection in acc. with standard	IEC 60947-7-1
Nominal current I_N	24 A
Nominal voltage U_N	1000 V
Open side panel	ja

Connection data

Conductor cross section solid min.	0.14 mm ²
Conductor cross section solid max.	4 mm ²
Conductor cross section stranded min.	0.14 mm ²
Conductor cross section stranded max.	4 mm ²
Conductor cross section AWG/kcmil min.	26
Conductor cross section AWG/kcmil max	12
Conductor cross section stranded, with ferrule without plastic sleeve min.	0.14 mm ²
Conductor cross section stranded, with ferrule without plastic sleeve max.	2.5 mm ²
Conductor cross section stranded, with ferrule with plastic sleeve min.	0.14 mm ²
Conductor cross section stranded, with ferrule with plastic sleeve max.	2.5 mm ²
2 conductors with same cross section, solid min.	0.14 mm ²
2 conductors with same cross section, solid max.	1.5 mm ²
2 conductors with same cross section, stranded min.	0.14 mm ²

2 conductors with same cross section, stranded max.	1.5 mm ²
2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, min.	0.5 mm ²
2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, max.	1.5 mm ²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, min.	0.14 mm ²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, max.	1.5 mm ²
Type of connection	Screw connection
Stripping length	9 mm
Internal cylindrical gage	A3
Screw thread	M3
Tightening torque, min	0.6 Nm
Tightening torque max	0.8 Nm

Certificates / Approvals



Certification

CB, CSA, CUL, DNV, GL, GOST, LR, UL, VDE-PZI

Certification Ex:

IECEX, KEMA-EX

CSA

Nominal voltage U _N	600 V
Nominal current I _N	20 A
AWG/kcmil	26-12

CUL

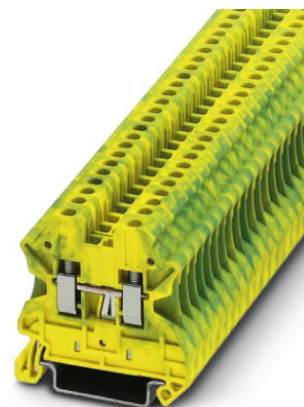
Nominal voltage U _N	600 V
Nominal current I _N	20 A
AWG/kcmil	26-12

UL

Nominal voltage U _N	600 V
Nominal current I _N	20 A
AWG/kcmil	26-12

UT 2,5-PE

Order No.: 3044092

<http://eshop.phoenixcontact.de/phoenix/treeViewClick.do?UID=3044092>

Feed-through modular terminal block, Type of connection: Screw connection, Screw connection, Cross section: 0.14 mm² - 4 mm², AWG 26 - 12, Width: 5.2 mm, Color: green-yellow, Mounting type: NS 35/7,5, NS 35/15



Commercial data

EAN	4017918960360
Pack	50 pcs.
Customs tariff	85369010
Weight/Piece	0.012156 KG
Catalog page information	Page 32 (CL-2009)

Product notes

WEEE/RoHS-compliant since:
01/01/2003



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Technical data

General

Number of levels	1
Number of connections	2
Color	green-yellow

Insulating material	PA
Inflammability class acc. to UL 94	V0

Dimensions

Width	5.2 mm
Length	47.7 mm
Height NS 35/7,5	47.5 mm
Height NS 35/15	55 mm

Technical data

Rated surge voltage	8 kV
Pollution degree	3
Surge voltage category	III
Insulating material group	I
Connection in acc. with standard	IEC 60947-7-2
Open side panel	ja

Connection data

Conductor cross section solid min.	0.14 mm ²
Conductor cross section solid max.	4 mm ²
Conductor cross section stranded min.	0.14 mm ²
Conductor cross section stranded max.	4 mm ²
Conductor cross section AWG/kcmil min.	26
Conductor cross section AWG/kcmil max	12
Conductor cross section stranded, with ferrule without plastic sleeve min.	0.14 mm ²
Conductor cross section stranded, with ferrule without plastic sleeve max.	2.5 mm ²
Conductor cross section stranded, with ferrule with plastic sleeve min.	0.14 mm ²
Conductor cross section stranded, with ferrule with plastic sleeve max.	2.5 mm ²
2 conductors with same cross section, solid min.	0.14 mm ²
2 conductors with same cross section, solid max.	1.5 mm ²
2 conductors with same cross section, stranded min.	0.14 mm ²
2 conductors with same cross section, stranded max.	1.5 mm ²

2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, min.	0.5 mm ²
2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, max.	1.5 mm ²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, min.	0.14 mm ²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, max.	1.5 mm ²
Type of connection	Screw connection
Stripping length	9 mm
Internal cylindrical gage	A3
Screw thread	M3
Tightening torque, min	0.6 Nm
Tightening torque max	0.8 Nm

Certificates / Approvals



Certification

CB, CSA, CUL, DNV, GL, LR, UL, VDE-PZI

Certification Ex:

IECEX, KEMA-EX

CSA

AWG/kcmil	26-12
-----------	-------

CUL

AWG/kcmil	26-12
-----------	-------

UL

AWG/kcmil	26-12
-----------	-------

D-UT 2,5/10

Order No.: 3047028



<http://eshop.phoenixcontact.de/phoenix/treeViewClick.do?UID=3047028>

End cover, Length: 47.7 mm, Width: 2.2 mm, Height: 48.4 mm,
Color: gray

Commercial data

EAN	4017918960346
Pack	50 pcs.
Customs tariff	85389099
Weight/Piece	0.00248 KG
Catalog page information	Page 26 (CL-2009)

Product notes

WEEE/RoHS-compliant since:
01/01/2003



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E/NS 35 N - End bracket

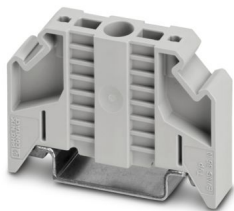
0800886

<https://www.phoenixcontact.com/us/products/0800886>



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End clamp, width: 9.5 mm, color: gray



Your advantages

- Large-surface marking

Commercial data

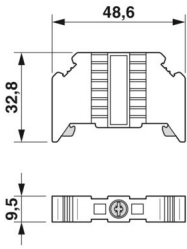
Item number	0800886
Packing unit	50 pc
Minimum order quantity	50 pc
Sales key	BE07
Product key	BE7111
GTIN	4017918129309
Weight per piece (including packing)	14.16 g
Weight per piece (excluding packing)	13.708 g
Customs tariff number	39269097
Country of origin	DE

Technical data

Connection data

Screw thread	M3
Tightening torque	0.5 ... 0.8 Nm

Dimensions

Dimensional drawing	
Width	9.5 mm
Height	48.6 mm
Depth	32.8 mm
Depth on NS 35/7,5	34.3 mm
Length	48.6 mm

Material specifications

Color	gray (RAL 7042)
Material	PA
Flammability rating according to UL 94	V0
Static insulating material application in cold	-60 °C
Temperature index of insulation material (DIN EN 60216-1 (VDE 0304-21))	125 °C
Relative insulation material temperature index (Elec., UL 746 B)	130 °C

Environmental and real-life conditions

Ambient conditions

Ambient temperature (operation)	-60 °C ... 105 °C (max. short-term operating temperature RTI Elec.)
Ambient temperature (storage/transport)	-25 °C ... 60 °C (for a short time, not exceeding 24 h, -60 °C to +70 °C)
Ambient temperature (assembly)	-5 °C ... 70 °C
Ambient temperature (actuation)	-5 °C ... 70 °C
Permissible humidity (storage/transport)	30 % ... 70 %

Mounting

Mounting type	NS 35/7,5
	NS 35/15

E/NS 35 N - End bracket

0800886

<https://www.phoenixcontact.com/us/products/0800886>



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info@phoenixcon.com

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Drawings Index:

Drawing Number	Drawing Description
D32384-40	ENCLOSURE EXT. LAYOUT
D32384-41	ENCLOSURE INT. LAYOUT
D32384-42	120VAC WIRING DIAGRAM
D32384-43	120VAC WIRING DIAGRAM

**32384 CHARLES COUNTY POMONKEY PUMP STATION
IMPROVEMENTS**

Drawings:

WARNING NAMEPLATE DETAIL

S-L CONTROLS, DWG D32384-40
VOLTAGE: 120VAC, 60HZ, SINGLE PHASE
AMPS: 0.369

SHORT CIRCUIT CURRENT: N/A

ENCLOSURE RATING: UL TYPE 4X

CAUTION: MULTIPLE POWER SOURCES ARE PROVIDED FROM FIELD
TERMINALS. EACH SOURCE SHOULD BE DISCONNECTED FOR
SERVICING.

WARNING: TO MAINTAIN OVERCURRENT, SHORT CIRCUIT, AND
GROUND FAULT PROTECTION, THE MANUFACTURER'S INSTRUCTIONS
FOR SELECTING CURRENT ELEMENTS MUST BE FOLLOWED

NAMEPLATES SIZED AS REQUIRED

NAME PLATE LEGEND TABLE

ITEM	QTY	DESCRIPTION
1	1	SECONDARY CONTROL PANEL
2	1	CONTROL POWER
3	1	SECONDARY CONTROL / ACTIVE
4	1	FAULT RESET
5	1	PUMP DOWN / FAILURE

ADDITIONAL PANEL LABELS:

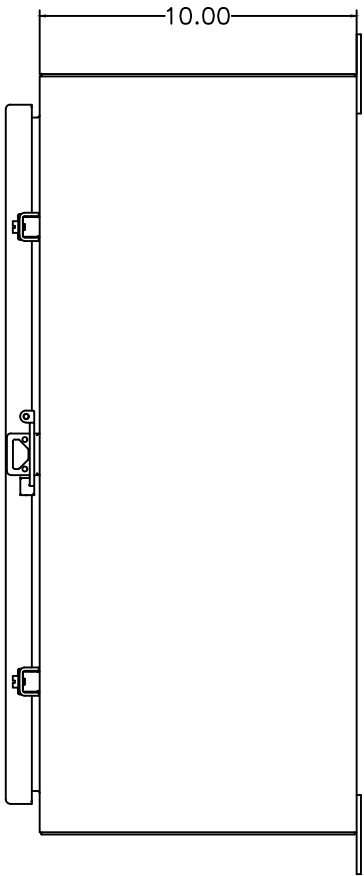
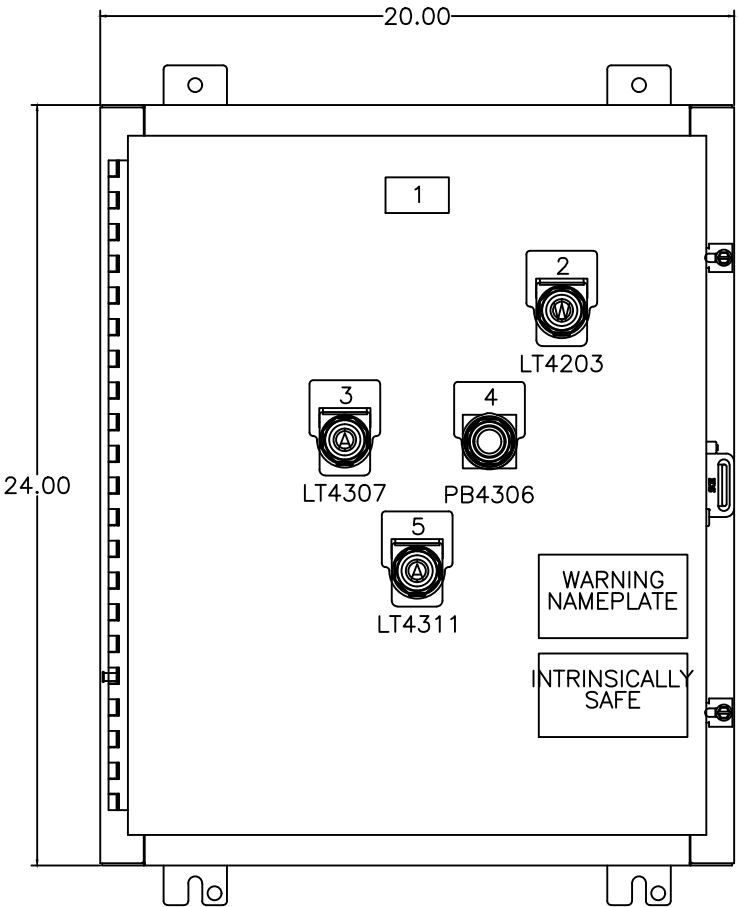
A* = "INTRINSICALLY SAFE FIELD WIRING TERMIANLS"
B* = "WARNING - EXPLOSION HAZARD - SUBSTITUTION OF
COMPONENTS MAY IMPAIR INTRINSIC SAFETY"
C* = "INTRINSICALLY SAFE CIRCUITS" (TO BE PLACED BY
CONTRACTOR AT CONDUIT OPENING)

NOTES:

ALL PLATES SHALL BE ENGRAVED ON WHITE BACKGROUND WITH
BLACK CHARACTERS EXCEPT WARNING NAMEPLATES WHICH SHALL
BE ENGRAVED ON RED BACKGROUND WITH WHITE CHARACTERS.
LETTER HEIGHT SHOULD BE MINIMUM 1/4".

WIRING:

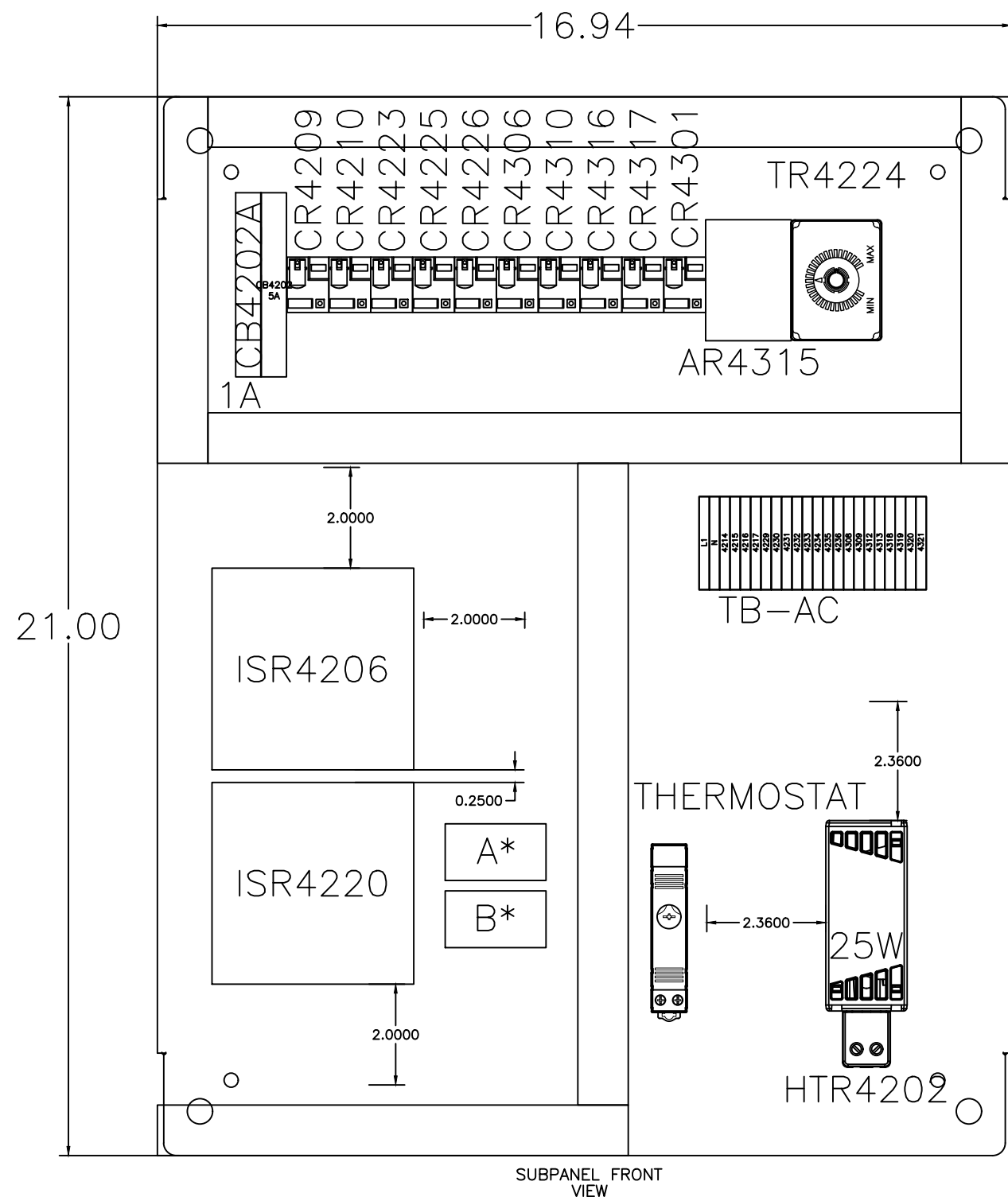
- 1) □ - INDICATES CABINET TERMINAL BLOCK
- 2) DASHED LINES INDICATE WIRING REMOTE TO CONTROL PANEL.
- 3) PROVIDE 20% SPARE TERMINALS
- 4) UL APPROVED NO. TO BE PROVIDED AT TIME OF MANUFACTURING
- 5) ELECTRICAL CONNECTION TORQUES:
TERMINAL: 12 INCH POUNDS
CIRCUIT BREAKER: 45 INCH POUNDS



PANEL WIRING

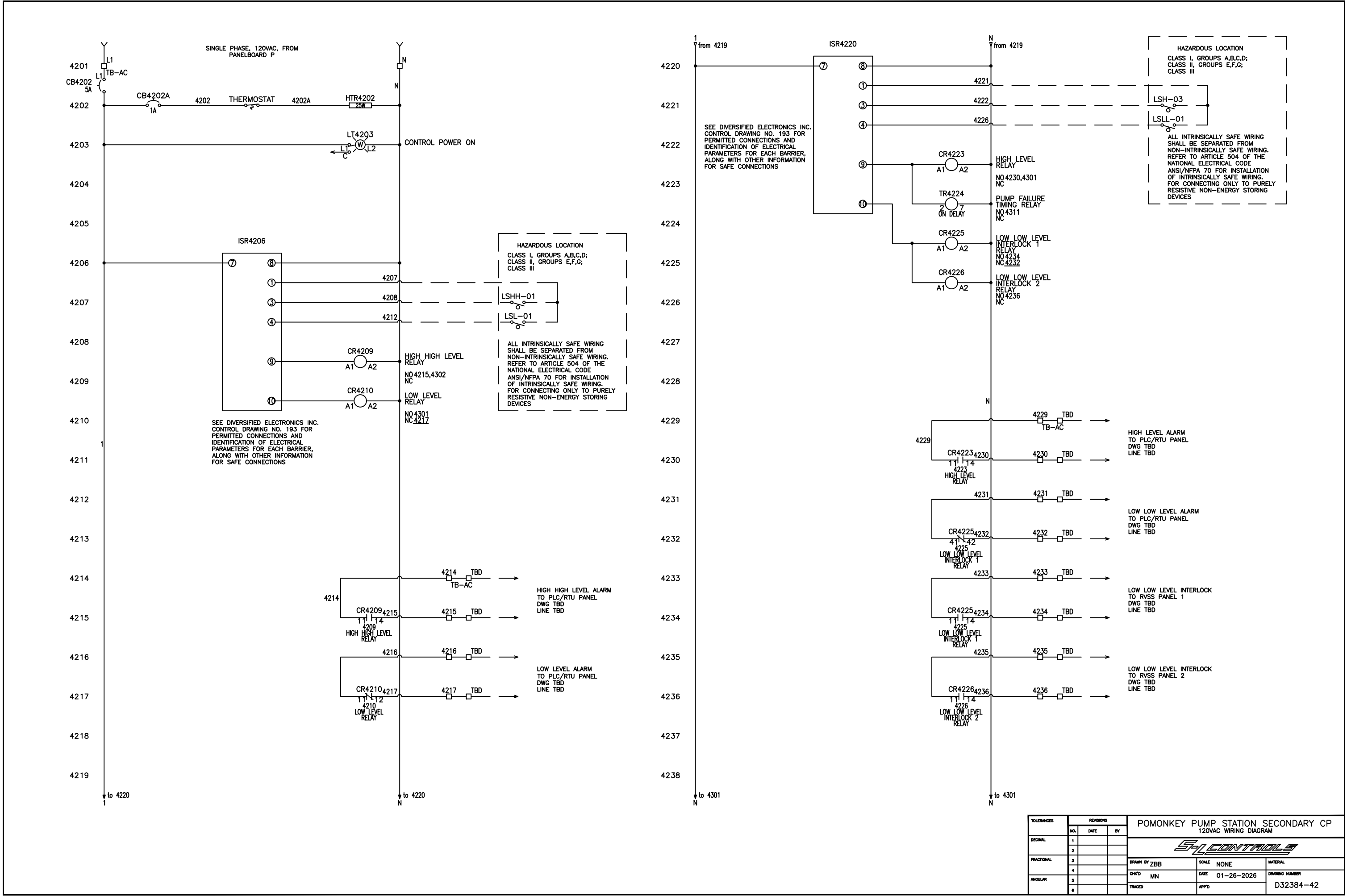
AREA	VOLTAGE	NOUN NAME	ID	GAGE	COLOR	TYPE
INTERNAL	120VAC	HOT	(L1), (1)	14	BLACK	MTW
INTERNAL	120VAC	NEUTRAL	(N)	14	WHITE	MTW
CONTROL	120VAC	HOT	VARIOUS	16	RED	MTW
CONTROL	120VAC	NEUTRAL	N	16	WHITE	MTW

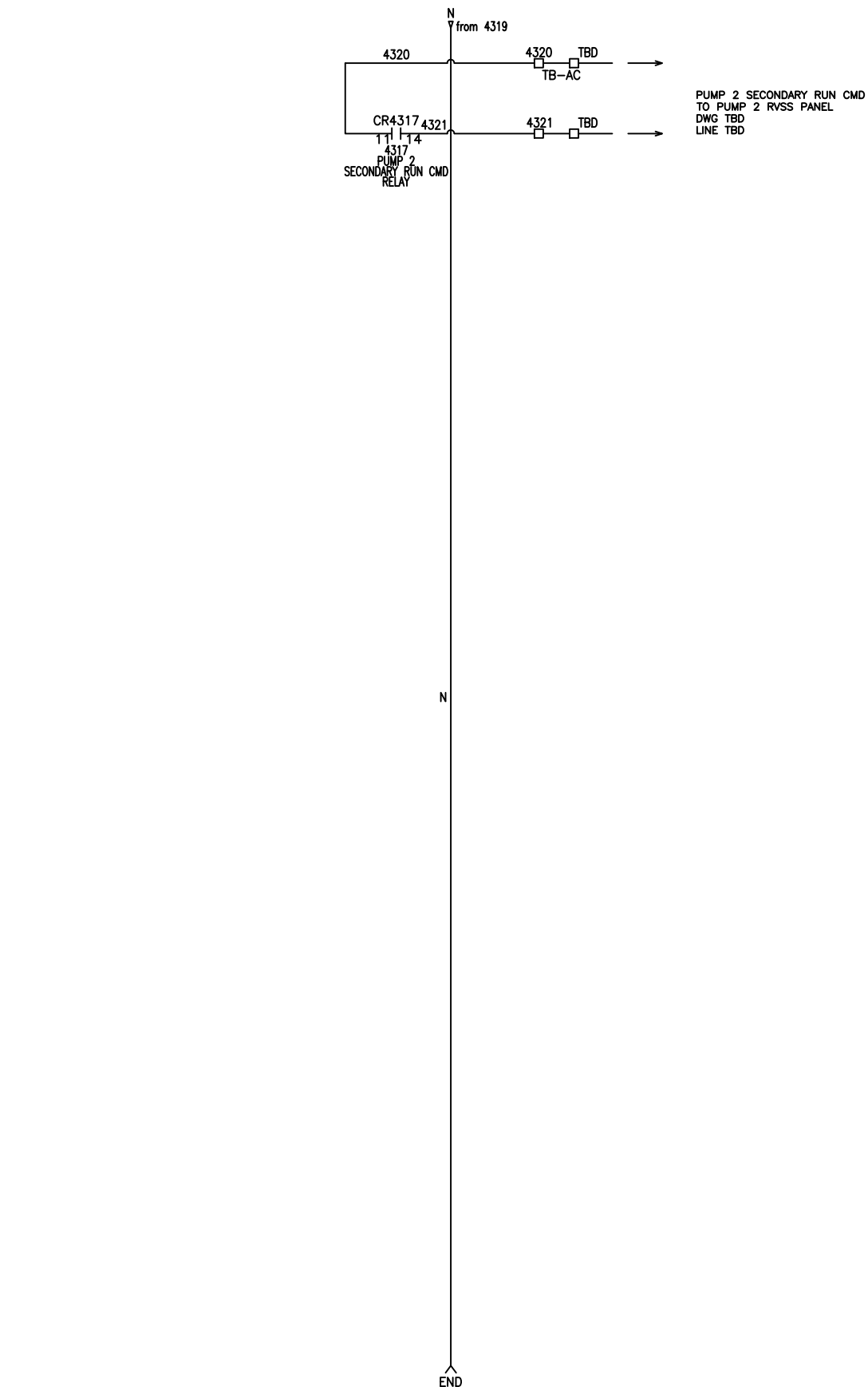
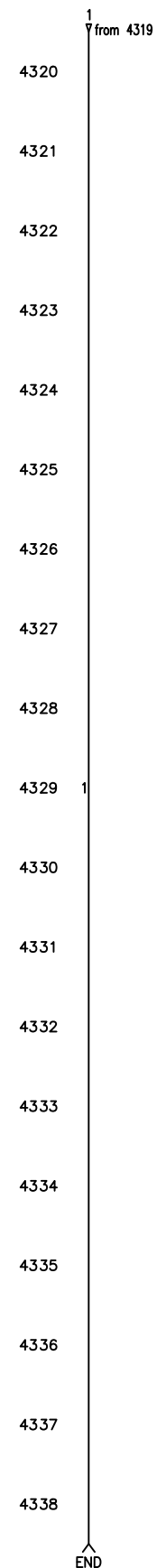
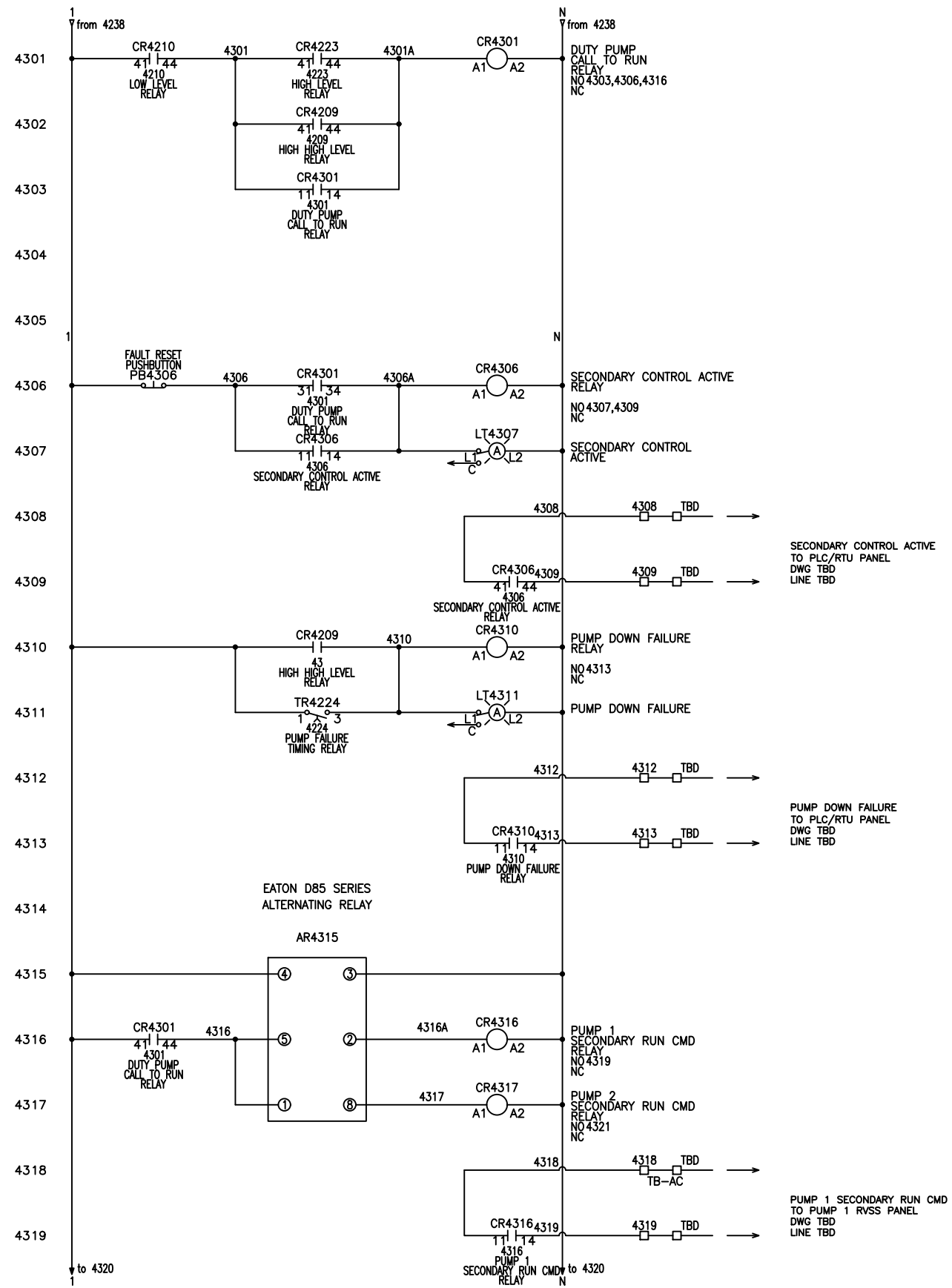
TOLERANCES	REVISIONS			POMONKEY PUMP STATION SECONDARY CP		
	NO.	DATE	BY	ENCLOSURE EXT. LAYOUT		
DECIMAL	1					
	2					
FRACTIONAL	3			DRAWN BY ZBB	SCALE NONE	MATERIAL
	4			CHK'D MN	DATE 01-26-2026	DRAWING NUMBER
ANGULAR	5			TRACED	APP'D	D32384-40
	6					



TAGS	QTY	CATALOG	DESCRIPTION	MANUFACTURER
ENC	1	SCE-24H2010SS6LP	WALL MOUNT ENCLOSURE – NEMA 4X	SCE
ENC	1	SCE-24P20	ENCLOSURE BACKPANEL	SCE
LT4203	1	9001SKT38LWW31	PTT LIGHT, 120VAC LED, WHITE	SCHNEIDER ELECTRIC
LT4307, LT4311	2	9001SKT38LYA31	PTT LIGHT, 120VAC LED, AMBER	SCHNEIDER ELECTRIC
PB4306	1	9001SKR1B	PUSHBUTTON, BLACK	SCHNEIDER ELECTRIC
PB4306	1	9001KA3	1 NC CONTACT	SCHNEIDER ELECTRIC
CB4202	1	7940102575	CIRCUIT BREAKER, 120VAC, 5A	WEIDMULLER
CB4202A	1	7940125379	CIRCUIT BREAKER, 120VAC, 1A	WEIDMULLER
CR4301	1	RXM3AB2F7	MINIATURE RELAY, 120VAC, 11 BLADE	SCHNEIDER ELECTRIC
CR4301	1	RXZE2S111M	MINIATURE RELAY SOCKET, 120VAC, DIN	SCHNEIDER ELECTRIC
CR4209, CR4210, CR4223, CR4225, CR4226, CR4306, CR4310, CR4316, CR4317	9	RXM2AB2F7	MINIATURE RELAY, 120VAC, 8 BLADE	SCHNEIDER ELECTRIC
CR4209, CR4210, CR4223, CR4225, CR4226, CR4306, CR4310, CR4316, CR4317	9	RXZE2M114	MINIATURE RELAY SOCKET, 120VAC, DIN	SCHNEIDER ELECTRIC
TR4224	1	9050JCK16V20	TIMING RELAY, 0–10 MIN, 120VAC, 8 PIN OCTAL, DPDT	SQUARE D
TR4224	1	8501NR1	TIMING RELAY, 120VAC, DIN	SQUARE D
ISR4206, ISR4220	2	ISO-120-AAE	INTRINSICALLY SAFE RELAY	DIVERSIFIED
AR4315	1	D851LA	ALTERNATING RELAY	EATON
AR4315	1	D3PA2	ALTERNATING RELAY SOCKET	EATON
HTR4202	1	SCE-TSH25	ENCLOSURE HEATER	SCE
THERMOSTAT	1	7T.81.0.000.2403	ENCLOSURE HEATER THERMOSTAT, NC	FINDER
TB-AC	22	3044076	UNIVERSAL TERMINAL BLOCK – UT 2,5	PHOENIX CONTACT
TB-AC	A/R	3047028	TERMINAL END BARRIER	PHOENIX CONTACT
TB-AC	A/R	3044092	GROUNDING TERMINAL BLOCK	PHOENIX CONTACT
TB-AC	A/R	0800886	TERMINAL ANCHOR	PHOENIX CONTACT

TOLERANCES		REVISIONS			POMONKEY PUMP STATION SECONDARY CP ENCLOSURE INT. LAYOUT		
		NO.	DATE	BY			
DECIMAL	1						
	2						
FRACTIONAL	3				DRAWN BY ZBB	SCALE NONE	MATERIAL
	4				CHK'D MN	DATE 01-26-2026	DRAWING NUMBER
ANGULAR	5				TRACED	APP'D	D32384-41
	6						





TOLERANCES		REVISIONS			POMONKEY PUMP STATION SECONDARY CP 120VAC WIRING DIAGRAM		
DECIMAL	NO.	DATE	BY				
	1						
FRACTIONAL	2						
	3						
ANGULAR	4				DRAWN BY ZBB	SCALE NONE	MATERIAL
	5				CHK'D MN	DATE 01-26-2026	DRAWING NUMBER
	6				TRACED	APP'D	D32384-43